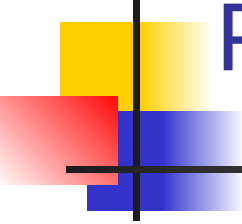


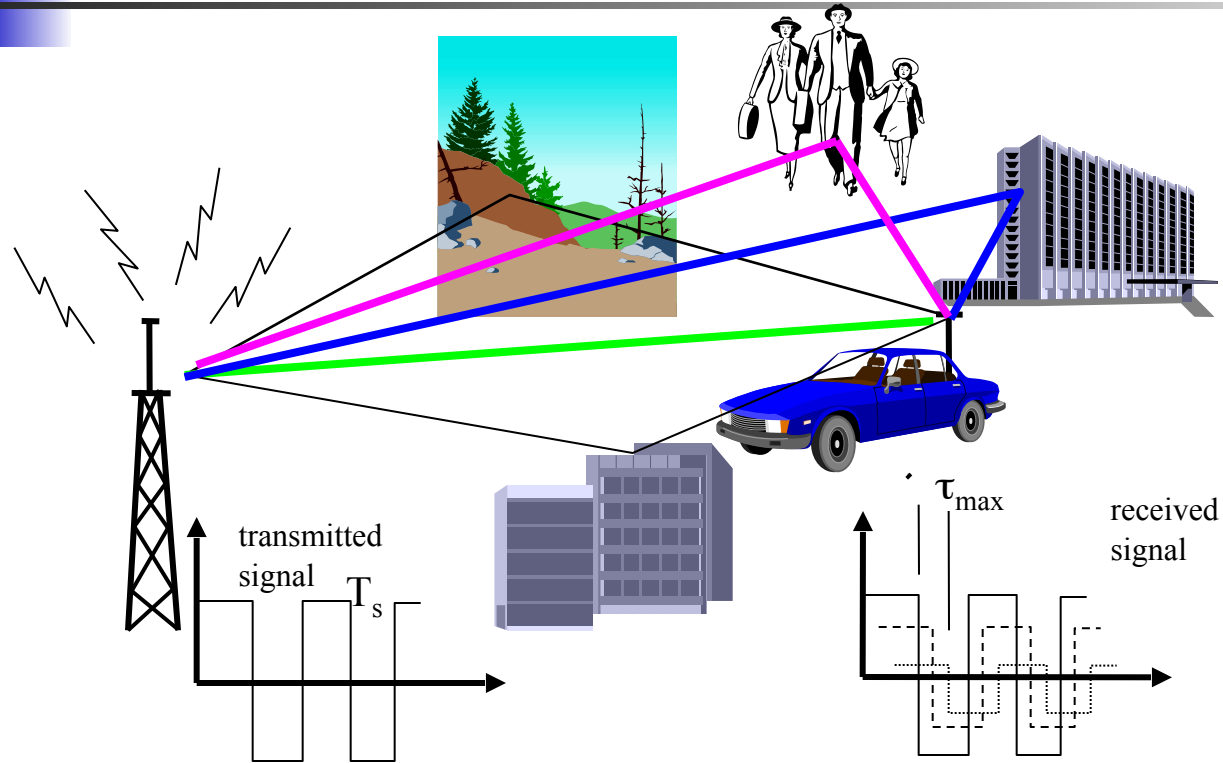
Propagation Through Channel



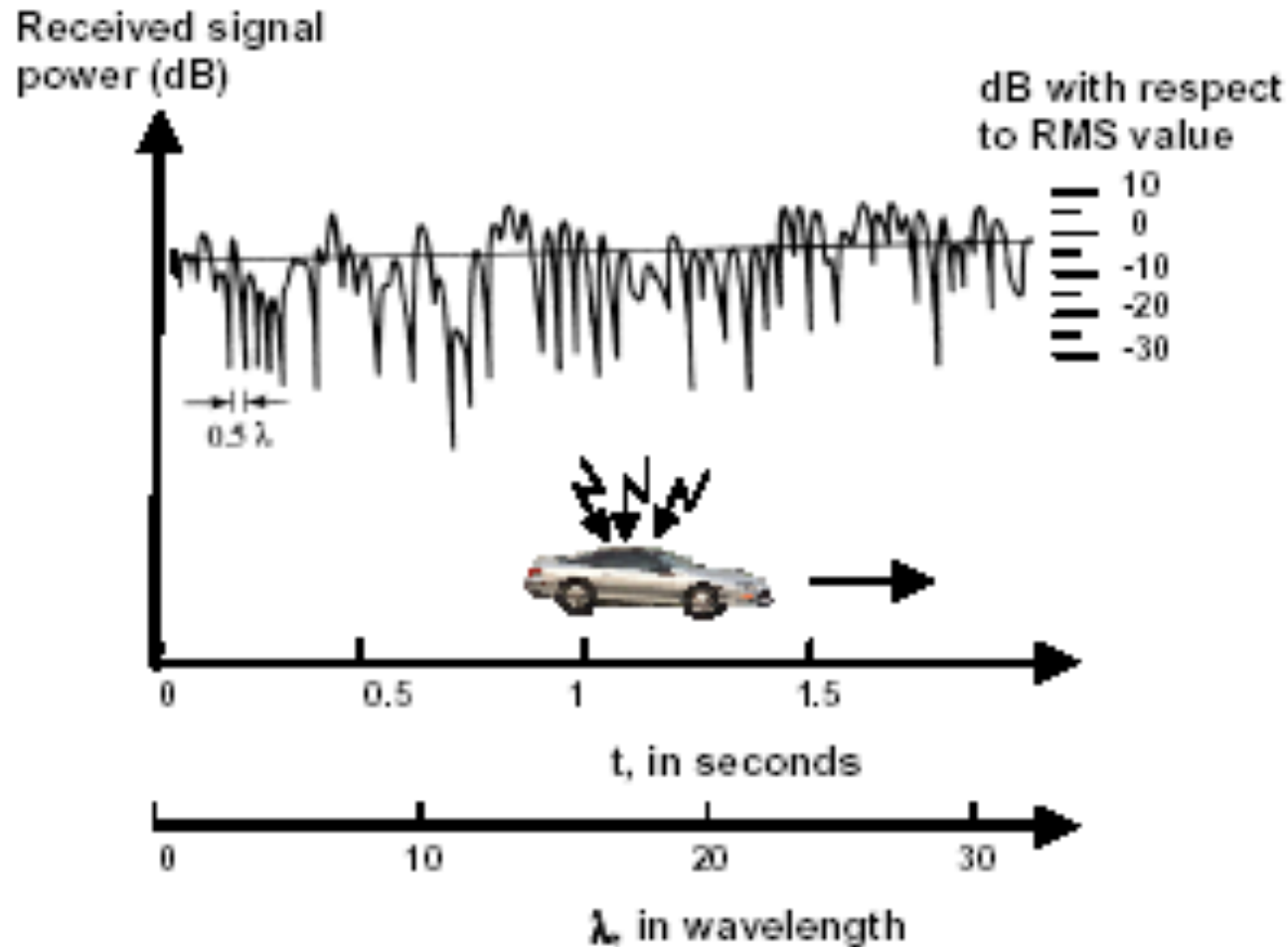
Radio Wave Propagation

- Reflection
 - Large buildings, earth surface
- Diffraction
 - Obstacles with dimensions in order of λ
- Scattering
 - Obstacles with size in the order of the wavelength of the signal or less
 - Foliage, lamp posts, street signs, walking pedestrian, etc.

Propagation Illustration

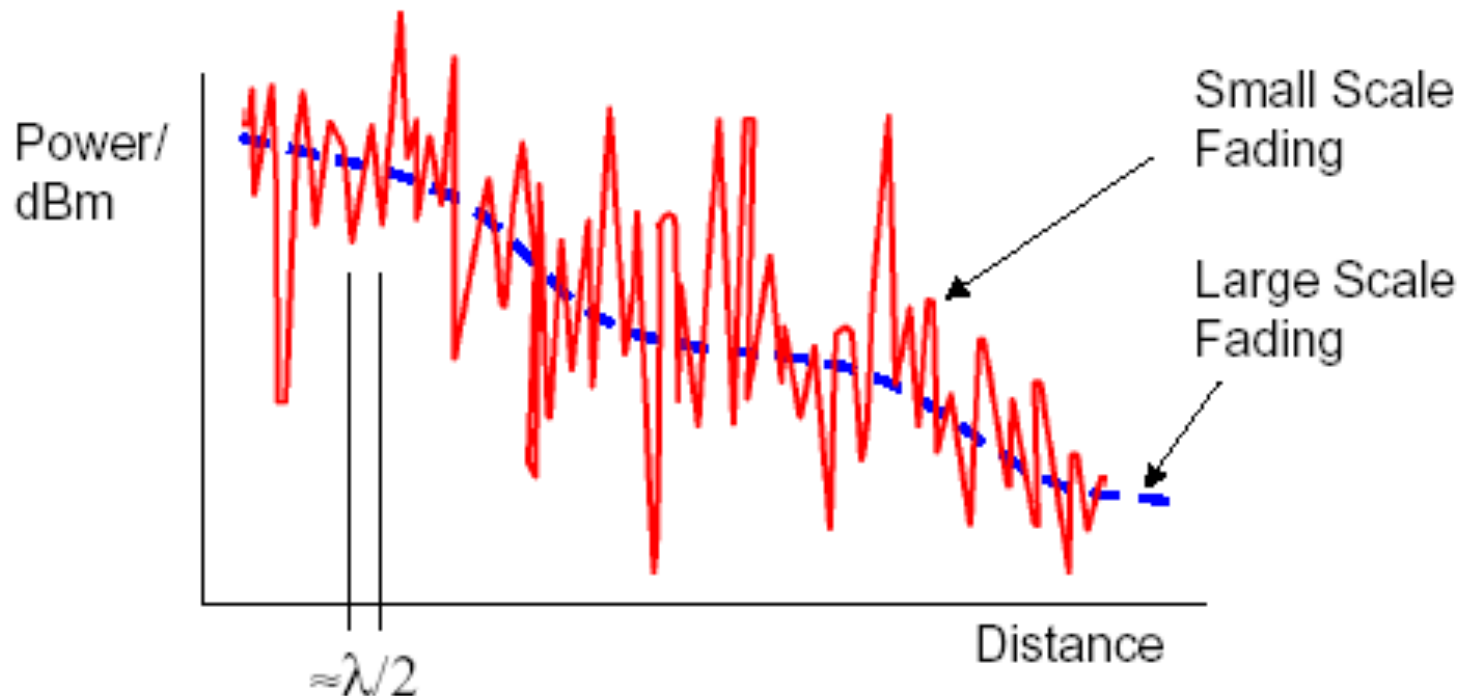


Large-Scale & Small-Scale Fading

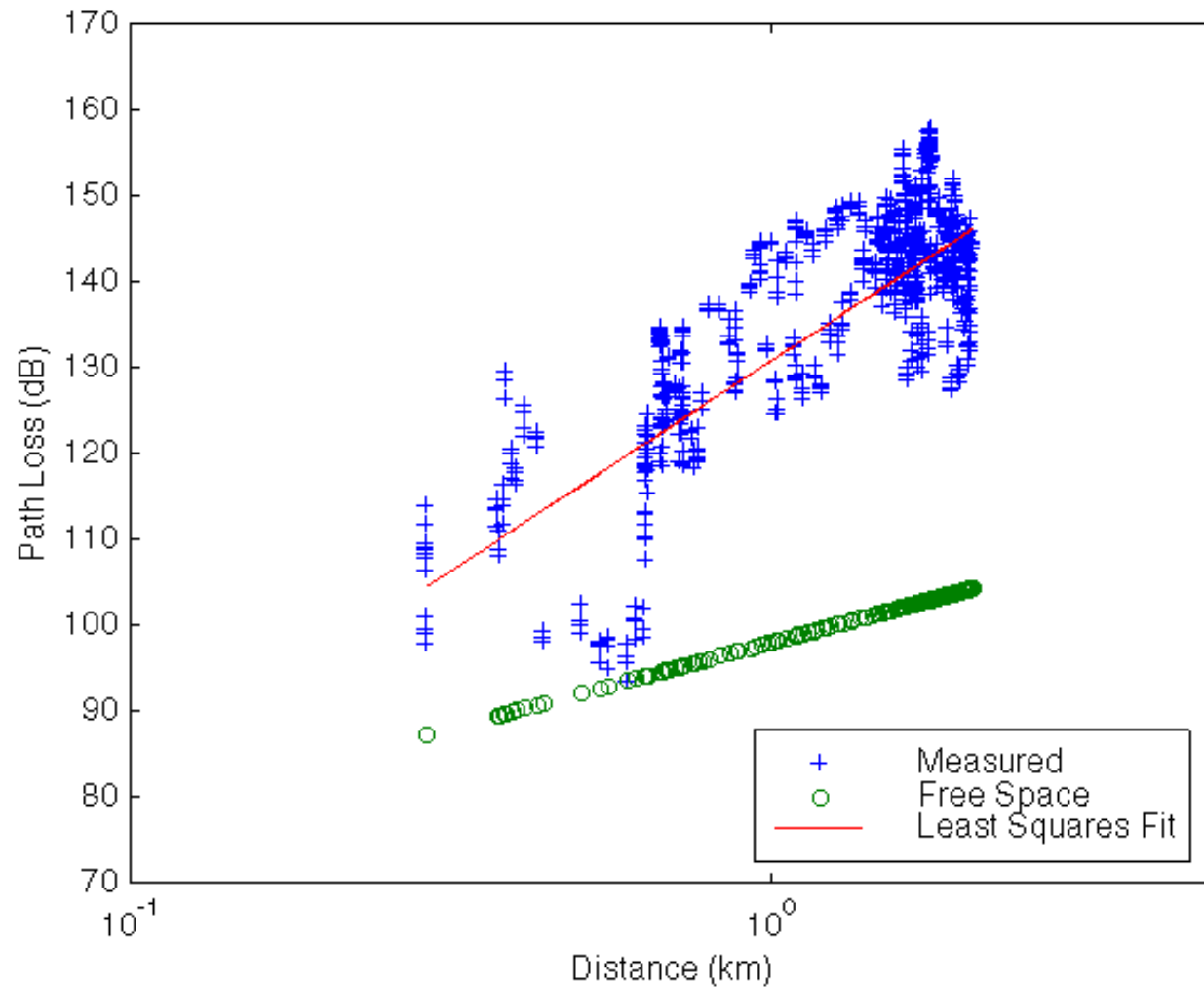


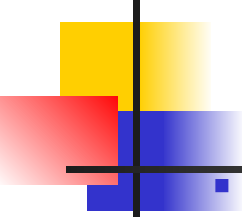
Large-Scale & Small-Scale Fading (Contd.)

The distance between small scale fades is on the order of $\lambda/2$



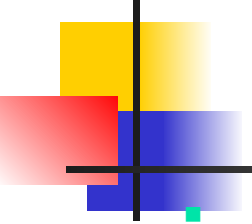
Path Loss





Propagation Models

- Usually, Maxwell's equations are Too complex to model the propagation.
- **Propagation Models** are normally used to predict the average signal strength at a given distance from the transmitter.
 - Propagation models that predict the mean signal strength for an arbitrary T-R separation distance are useful in estimating the radio coverage area. This is called the **Large Scale** or **Path Loss** propagation model (several hundreds or thousands of meters);
 - Propagation models that characterize the rapid fluctuations of the received signal strengths over very short distance (few wavelengths) or short duration (few seconds) are called **Small Scale** or **Fading** models.



Propagation Models (Contd.)

- Free Space Propagation Model - LOS path exists between T-R
- May applicable for satellite communication or microwave LOS links
- Frii's free space equation:

$$Pr(d) = Pt Gt Gr \lambda^2 / (4\pi)^2 d^2 L$$

- Pt : Transmitted power
 - Pr : Received power
 - Gt : Transmitter gain
 - Gr : Receiver gain
 - d : Distance of T-R separation
 - L : System loss factor
 - λ : Wavelength in meter
- **Path Loss** – difference (in dB) between the effective transmitted power and the received power

Propagation Models (Contd.)

- Modified free space equation

$$Pr(d) = Pr(d_0)(d_0/d)^2$$

- Modified free space equation in dB form

$$Pr(d) \text{ dBm} = 10 \log[Pr(d_0)/0.001W] + 20 \log(d_0/d)$$

$$\text{where } d \geq d_0 \geq d_f$$

- d_f is Fraunhofer distance which complies:

$$d_f = 2D^2/\lambda$$

where D is the largest physical linear dimension of the antenna

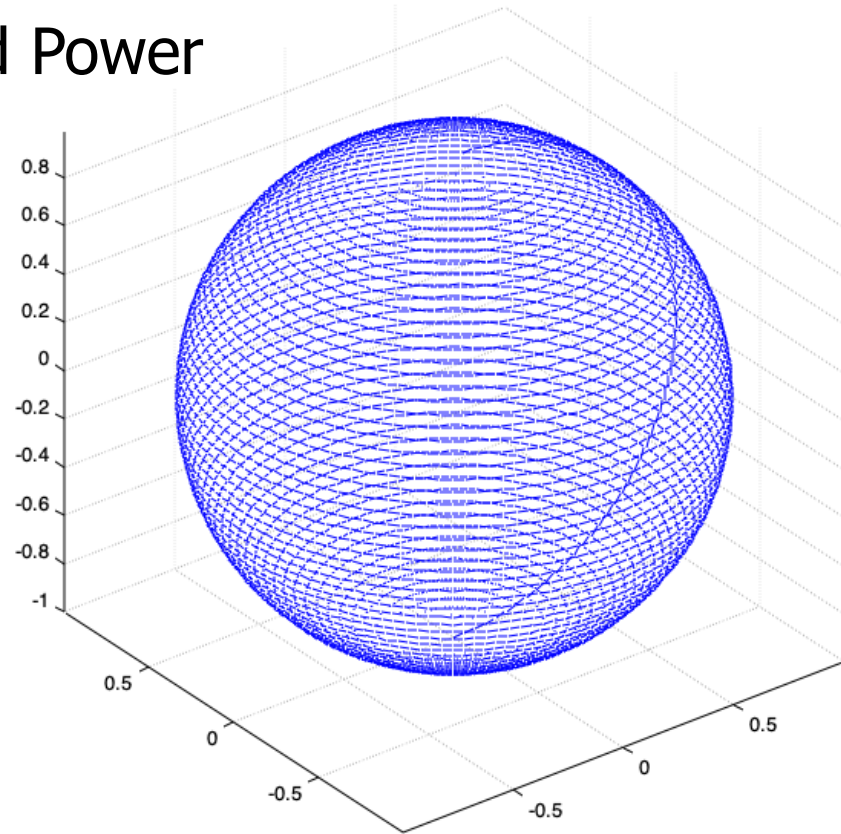
- In practice, reference distance is chosen to be 1m

EIRP

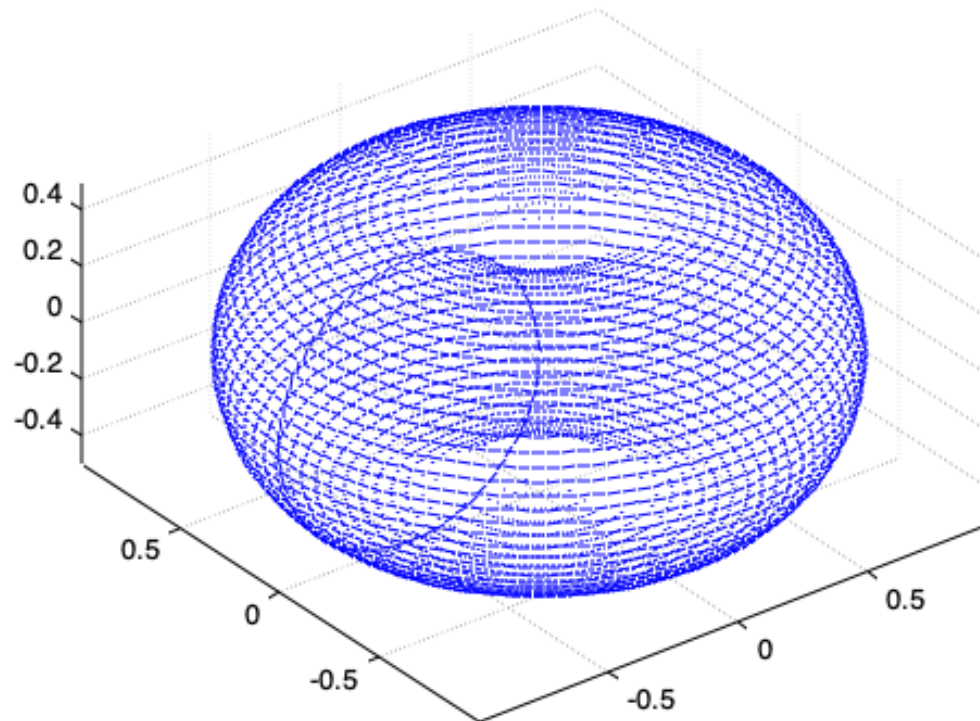
Effective Isotropic Radiated Power

$$\text{EIRP} = P_t G_t$$

which represents the maximum radiated power available from a transmitter in the direction of maximum antenna gain, as compared to an isotropic radiator.



ERP



In practice, effective radiated power (ERP) is used to denote the maximum radiated power as compared to a half-wave dipole antenna.

Link Budget

RF Link Budget Calculator

Free Space Loss Path

Frequency

0.9000

GHz

ERP

50.0000

Watts

ERP in dBm

46.9897

dBm

Transmission Line Loss

0.0000

dB

Tx Antenna Gain

0.0000

dBi

Path Length

0.1500

Km

Free Space Path Loss

75.0484

dB

Rx Antenna Gain

0.0000

dBi

Rx Transmission Line Loss

0.0000

dB

Rx Signal Strength

-28.0587

dBm

Rx Threshold

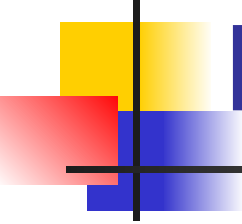
-85.0000

dBm

Fade Margin

56.9413

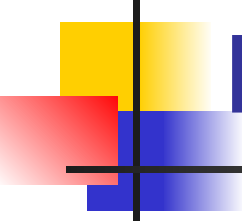
dB



Propagation Mechanisms

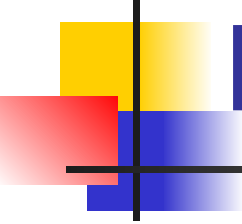
We next discuss propagation mechanisms (Reflection, Diffraction, and Scattering) because:

- They have an impact on the wave propagation in a mobile communication system
- The most important parameter, **Received power** is predicted by large scale propagation models based on the physics of reflection, diffraction and scattering



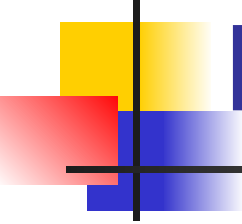
Reflection

- When a radio wave propagating in one medium impinges upon another medium having different electrical properties, the wave is partially reflected and partially transmitted
- **Fresnel Reflection Coefficient (Γ)** gives the relationship between the electric field intensity of the reflected and transmitted waves to the incident wave in the medium of origin
- **The Reflection Coefficient** is a function of the material properties, depending on
 - Wave Polarization
 - Angle of Incidence
 - Frequency of the propagating wave



Reflection

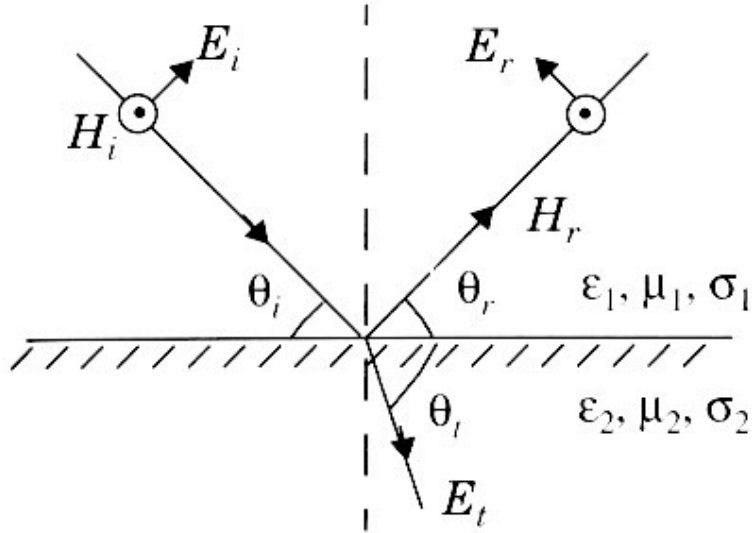
- **Reflection** occurs when a propagating electromagnetic wave impinges upon an object which has very **large dimensions when compared to the wavelength**, e.g., buildings, walls.
- Reflection occurs when RF energy is incident upon a boundary between two materials (e.g air/ground) with different electrical characteristics
 - Permittivity μ ,Permeability ϵ ,Conductance σ
- Reflecting surface must be large relative to λ of RF energy



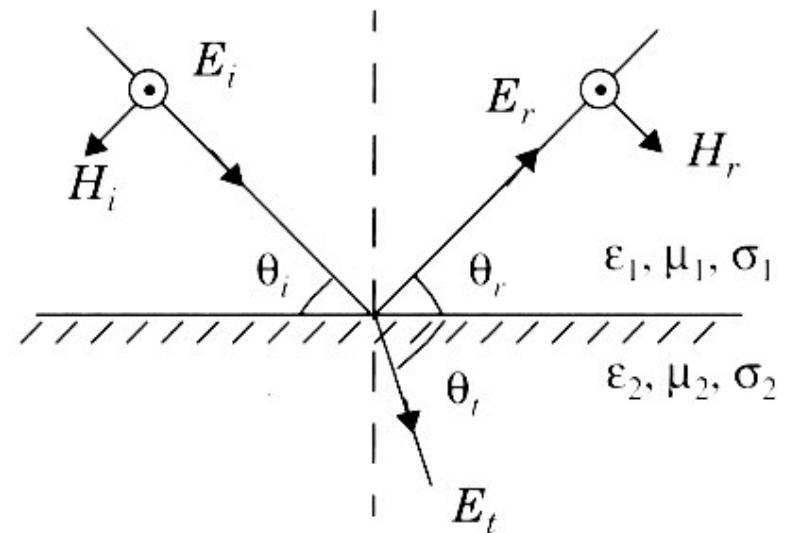
Reflection

- Perfect conductors reflect with no attenuation
 - Like light to the mirror
- Dielectrics reflect a fraction of incident energy
- Reflection coefficient of ground
 - (a) vertical polarization (v) or E field in the plane of incidence.
 - (b) horizontal polarization (h) or E field perpendicular to the incident plane

Reflection from Dielectric smooth surface



(a) E-field in the plane of incidence



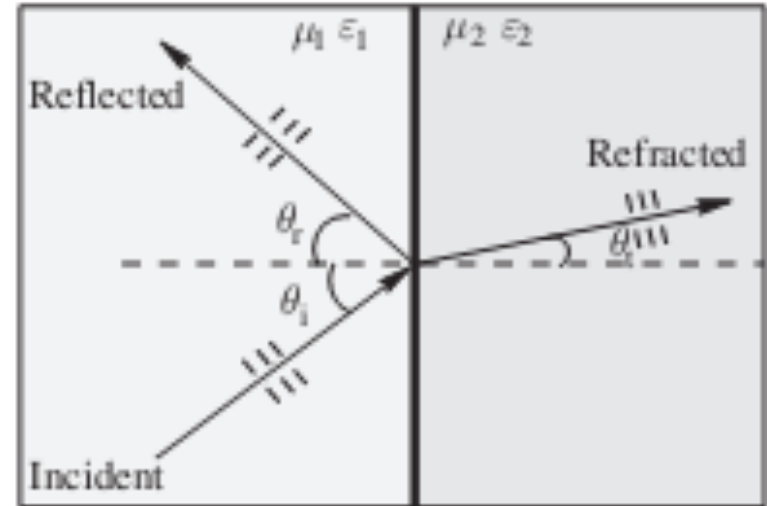
(b) E-field normal to the plane of incidence

Figure 4.4 Geometry for calculating the reflection coefficients between two dielectrics.

$$\epsilon = \epsilon_0 \epsilon_r - j\epsilon' \quad \text{For lossy dielectric material}$$

$$\epsilon = \epsilon_0 \epsilon_r \quad \text{For perfect dielectric}$$

Reflection

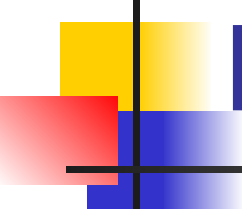


Snell's law : $\theta_i = \theta_r, \frac{\sin(90 - \theta_i)}{\sin(90 - \theta_t)} = \frac{n_2}{n_1}$

Boundary conditions :

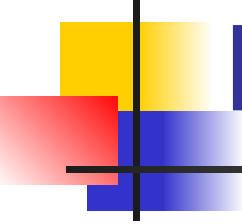
$$[E_1]_{//} = [E_2]_{//} \quad [H_1]_{//} = [H_2]_{//}$$

$$[\epsilon_1 E_1]_{\perp} = [\epsilon_2 E_2]_{\perp} \quad [\mu_1 H_1]_{\perp} = [\mu_2 H_2]_{\perp}$$



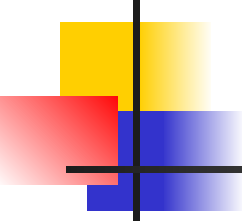
Reflection

If medium 1 is free space ($\varepsilon = \varepsilon_0$) and medium 2 is an dielectric, then $\Gamma_h \rightarrow -1, \Gamma_v \rightarrow -1$ regardless of ε_r if $\theta_i \rightarrow 0^\circ$. Ground may be modeled as a perfect reflector (reflection coefficient of unit magnitude) when an incident wave grazes the earth, regardless of polarization or ground dielectric properties.



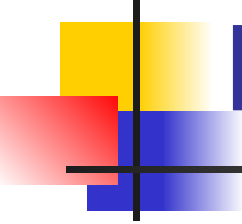
Brewster Angle

- It is the angle at which no reflection occurs in the medium of origin. It occurs when the incident angle θ_B is such that the reflection coefficient Γ_{II} is equal to zero. And
- $$\sin\theta_B = \sqrt{\frac{\epsilon_1}{\epsilon_1 + \epsilon_2}}$$
- It is only for vertical polarization.



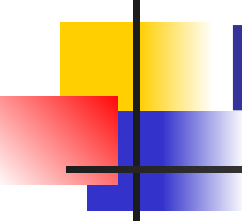
Ground Reflection (2- ray) Model

- In a mobile radio channel, a single direct path between the base station and mobile is rarely the only physical path for propagation
- Hence the free space propagation model in most cases is inaccurate when used alone
- The **2- ray GRM** is based on geometric optics
- It considers both- direct path and ground reflected propagation path between transmitter and receiver
- This was found reasonably accurate for predicting large scale signal strength over distances of several kilometers for mobile radio systems using tall towers (heights above 50 m), and also for **L-O-S** micro cell channels in urban environments



Diffraction

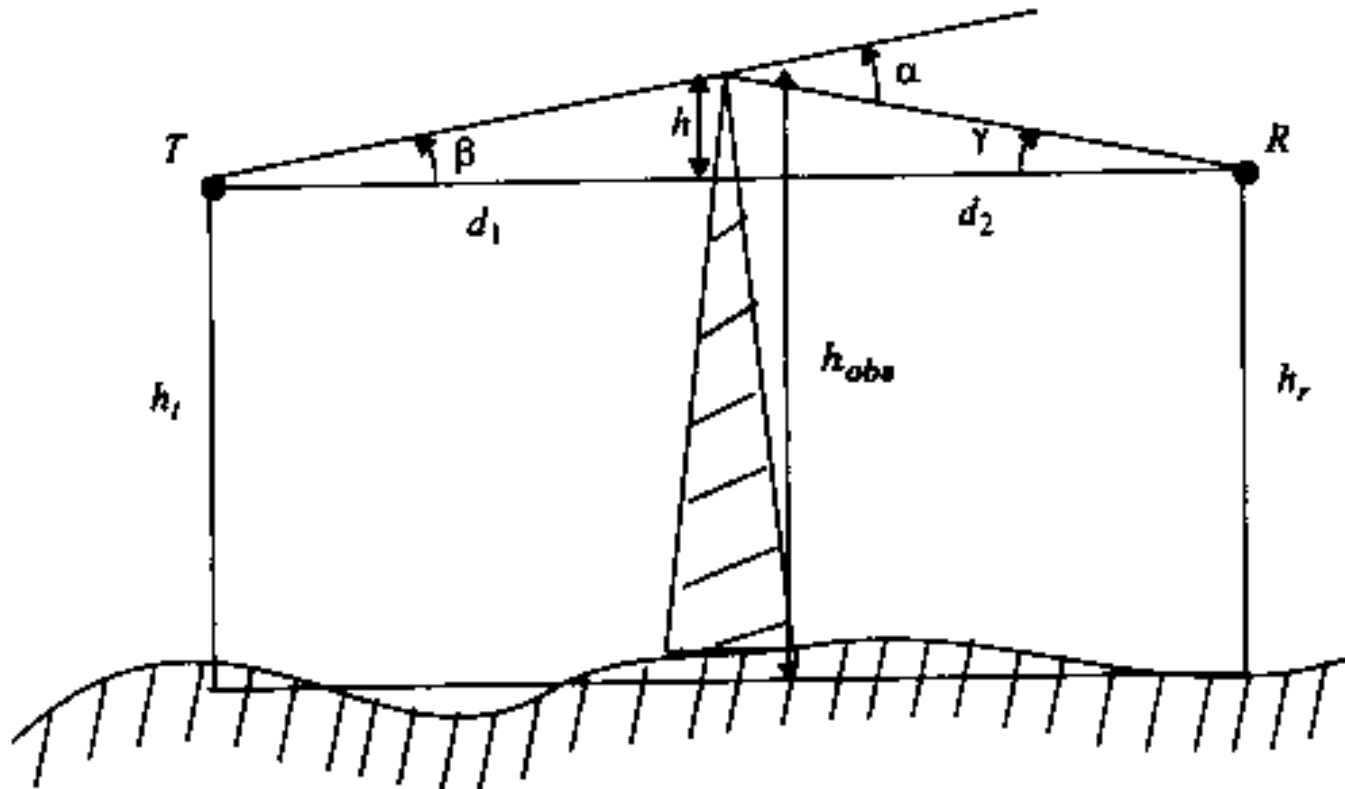
- Phenomena: Radio signal can propagate around the curved surface of the earth, beyond the horizon and behind obstructions.
- Radio path between transmitter and receiver obstructed by surface with sharp irregular edges.
- Although the received field strength decreases rapidly as a receiver moves deeper into the obstructed (shadowed) region, the diffraction field still exists and often has sufficient strength to produce a useful signal.
- The field strength of a diffracted wave in the shadowed region is the vector sum of the electric field components of all the secondary wavelets in the space around the obstacles.



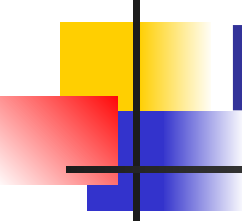
Knife-edge Diffraction Model

- It is essential to estimate the signal attenuation caused by diffraction of radio waves over hills and buildings in predicting the field strength in the given service area.
- In practice, prediction for diffraction loss is a process of theoretical approximation modified by necessary empirical corrections.
- The simplest case: shadowing is caused by a single object such as a hill or mountain.

Diffraction Geometry



(a) Knife-edge diffraction geometry. The point T denotes the transmitter and R denotes the receiver, with an infinite knife-edge obstruction blocking the line-of-sight path.



Parameters

- Fresnel-Kirchoff diffraction parameter

$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}} = \alpha \sqrt{\frac{2d_1 d_2}{\lambda(d_1 + d_2)}}$$

- The electric field strength E_d ,

$$\frac{E_d}{E_0} = F(v) = \frac{1+j}{2} \int_v^{\infty} \exp\left(-\frac{j\pi t^2}{2}\right) dt$$

where E_0 is the free space field strength

- The diffraction gain: $G_d(dB) = 20 \log |F(v)|$

Graphical representation

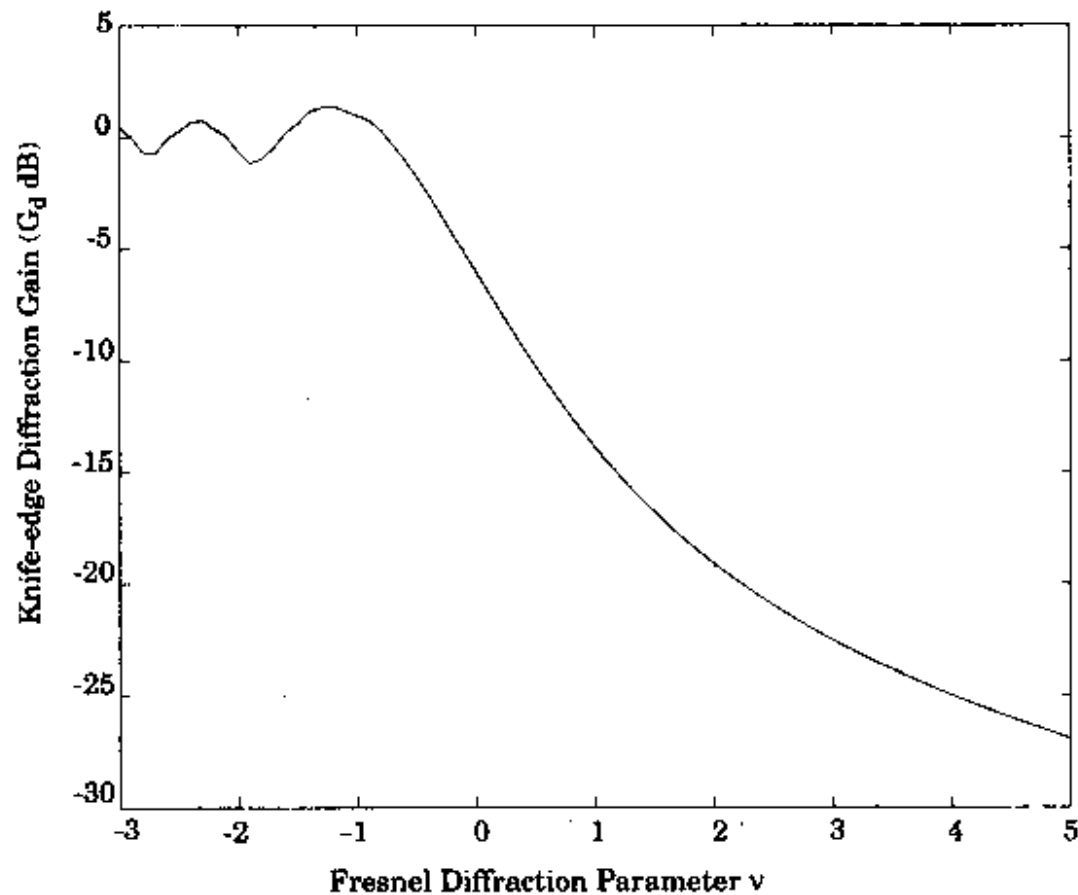
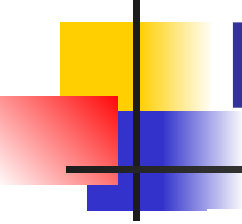


Figure 3.14
Knife-edge diffraction gain as a function of Fresnel diffraction parameter v .



Lee's Approximate

$$G_d(\text{dB}) = 0$$

$$v \leq -1$$

$$G_d(\text{dB}) = 20\log(0.5 - 0.62v)$$

$$-1 \leq v \leq 0$$

$$G_d(\text{dB}) = 20\log(0.5 \exp(-0.95v))$$

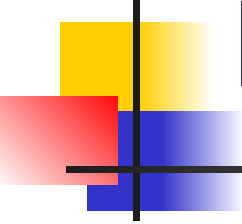
$$0 \leq v \leq 1$$

$$G_d(\text{dB}) = 20\log\left(0.4 - \sqrt{0.1184 - (0.38 - 0.1v)^2}\right)$$

$$1 \leq v \leq 2.4$$

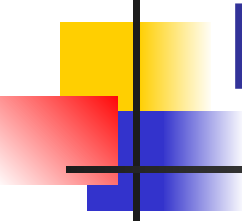
$$G_d(\text{dB}) = 20\log\left(\frac{0.225}{v}\right)$$

$$v > 2.4$$



Multiple Knife-edge Diffraction

- In the practical situations, especially in hilly terrain, the propagation path may consist of more than one obstruction.
- Optimistic solution (by Bullington): The series of obstacles are replaced by a single equivalent obstacle so that the path loss can be obtained using single knife-edge diffraction models.



Note

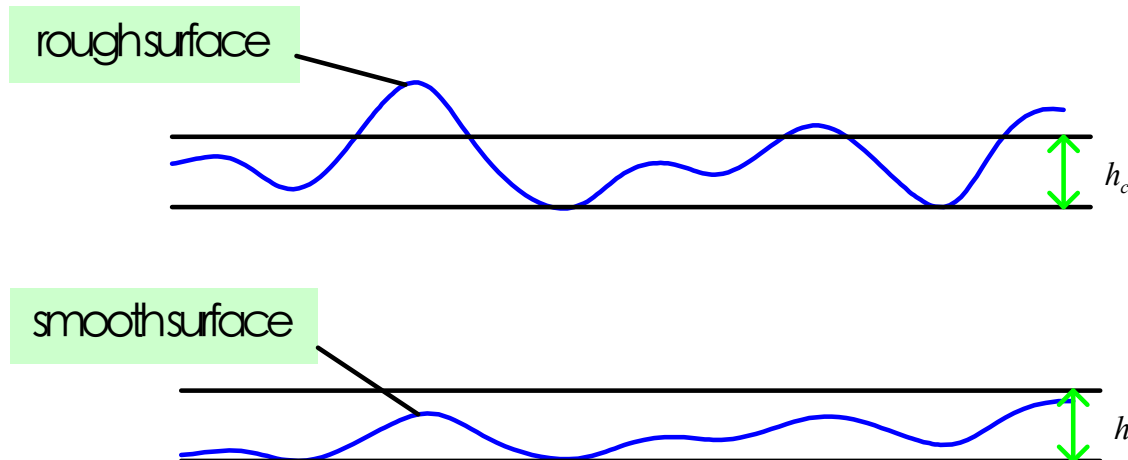
- The actual received signal in a mobile radio environment is often stronger than what is predicted by reflection and diffraction
- Reason:
When a radio wave impinges on a rough surface, the reflected energy is spread in all directions due to scattering

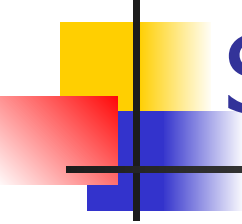
• Scattering

when a radio wave impinges on a rough surface, the reflected energy is spread out, e.g., trees, lamp posts.

- Surface roughness is test using Rayleigh criterion which defines a critical height h_c for a given angle of incidence θ_i

$$h_c = \frac{\lambda}{8 \sin \theta_i}$$





Scattering Loss Factor

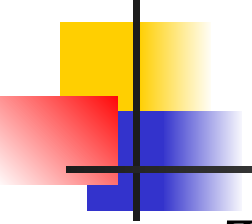
$$\rho_s = \exp[-8(\Pi\sigma_h \sin\theta_i)^2] I_0[8(\Pi\sigma_h \cos\theta_i)^2]$$

where ,

I_0 is the Bessel function of the first kind and zero order

σ_h is the standard deviation of the surface height, h about the mean surface height

θ_i is the angle of incidence

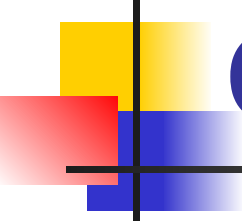


Radar cross section model:

- The radar cross section of a scattering object is defined as the ratio of the power density of the signal scattered in the direction of the receiver to the power density of the radio wave incident upon the scattering object, and has units of square meters.

Why do we require this?

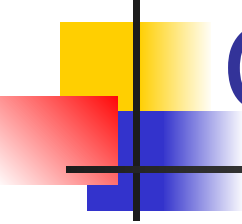
- In radio channels where large, distant objects induce scattering, the physical location of such objects can be used to accurately predict scattered signal strengths.



Continues

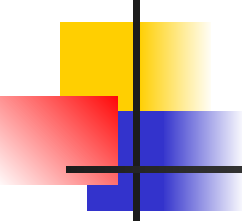
- For urban mobile radio systems ,models based on the bistatic radar equation is used to compute the received power due to scattering in the far field.
- The bistatic radar equation describes the propagation of a wave traveling in free space which impinges on a distant scattering object, and is the reradiated in the direction of the receiver, given by

$$P_R \text{ (dBm)} = P_T \text{ (dBm)} + G_T \text{ (dBi)} + 20\log(\lambda) + \text{RCS}[\text{dBm}^2] - 30\log(4\pi) - 20\log d_T - 20\log d_R$$



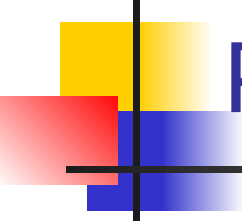
Continues

- Where d_T and d_R are the distance from the scattering object to the transmitter and receiver respectively.
- In the above equation the scattering object is assumed to be in the (far field) Fraunhofer region of both the transmitter and receiver and is useful for predicting receiver power which scatters off large objects such as buildings, which are for both the transmitter and receiver.



Path Loss Models

- Radio Propagation models are derived using a combination of empirical and analytical methods.
- These methods implicitly take into account all the propagation factors both known and unknown through the actual measurements.
- Path loss models are used to estimate the received signal level as a function of distance.
- With the help of this model we can predict SNR for a mobile communication system.

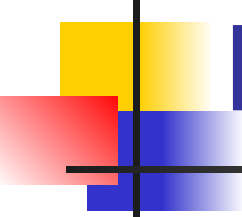


Path Loss Models (Cont'd)

- Two such models
 - **Log - Distance Path Loss Model**
 - **Log - Normal Shadowing**
- The path loss at a particular location for any value of d is random and distributed log-normally about the mean distance- dependent value is given by

$$PL(d)[dB] = PL(d) + X_{\sigma} = PL(d_0) + 10n \log(d/d_0) + X_{\sigma}$$

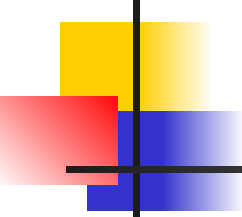
where, X_{σ} is the Zero –mean Gaussian distributed random variable with standard deviation σ (also in dB)



Path Loss Exponents

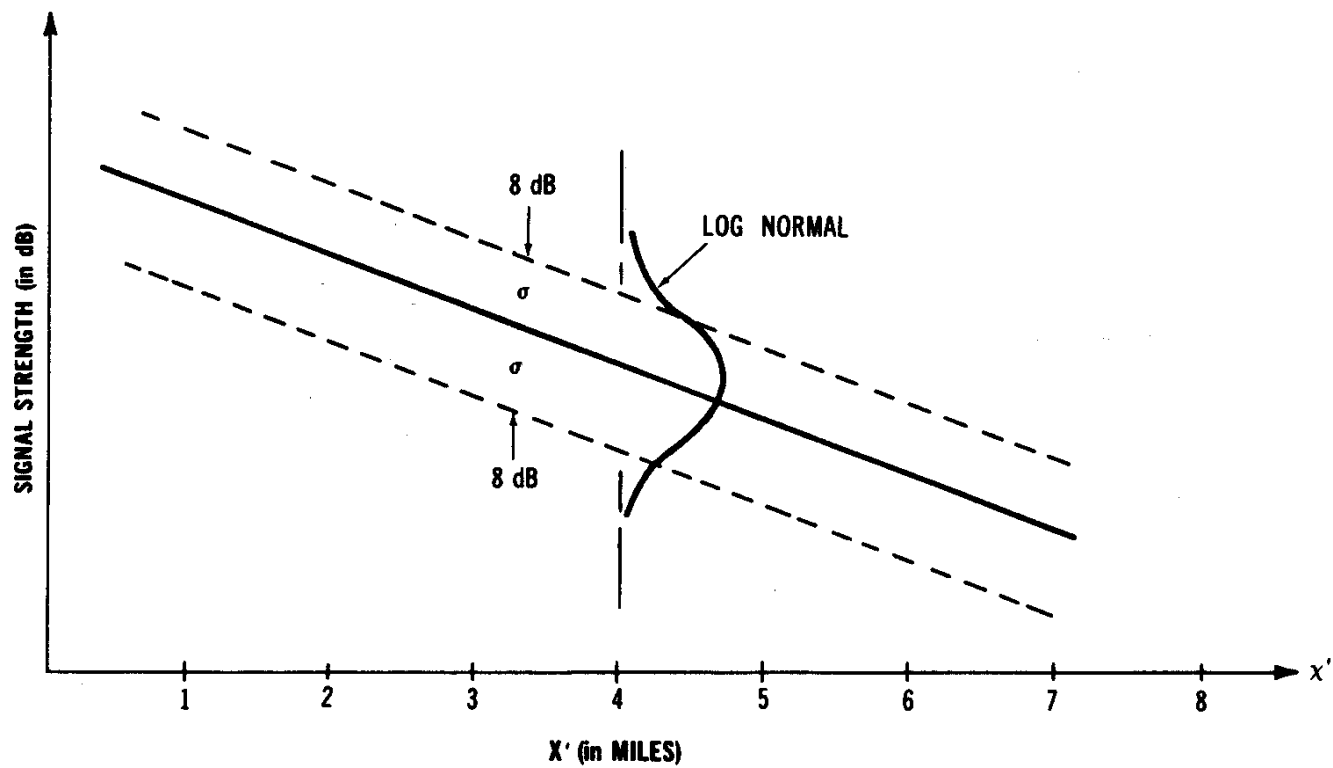
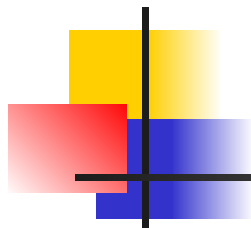
Table 3.2 Path Loss Exponents for Different Environments

Environment	Path Loss Exponent, n
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line-of-sight	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3



Log-Normal Distribution:

- It describes the random shadowing effects which occur over a large number of measurement locations which have the same T-R separation, but have different levels of clutter on the propagation path.
- The random effects of shadowing are accounted for using the Gaussian distribution
- In practice, the values of n and σ are often computed from measured data, using linear regression



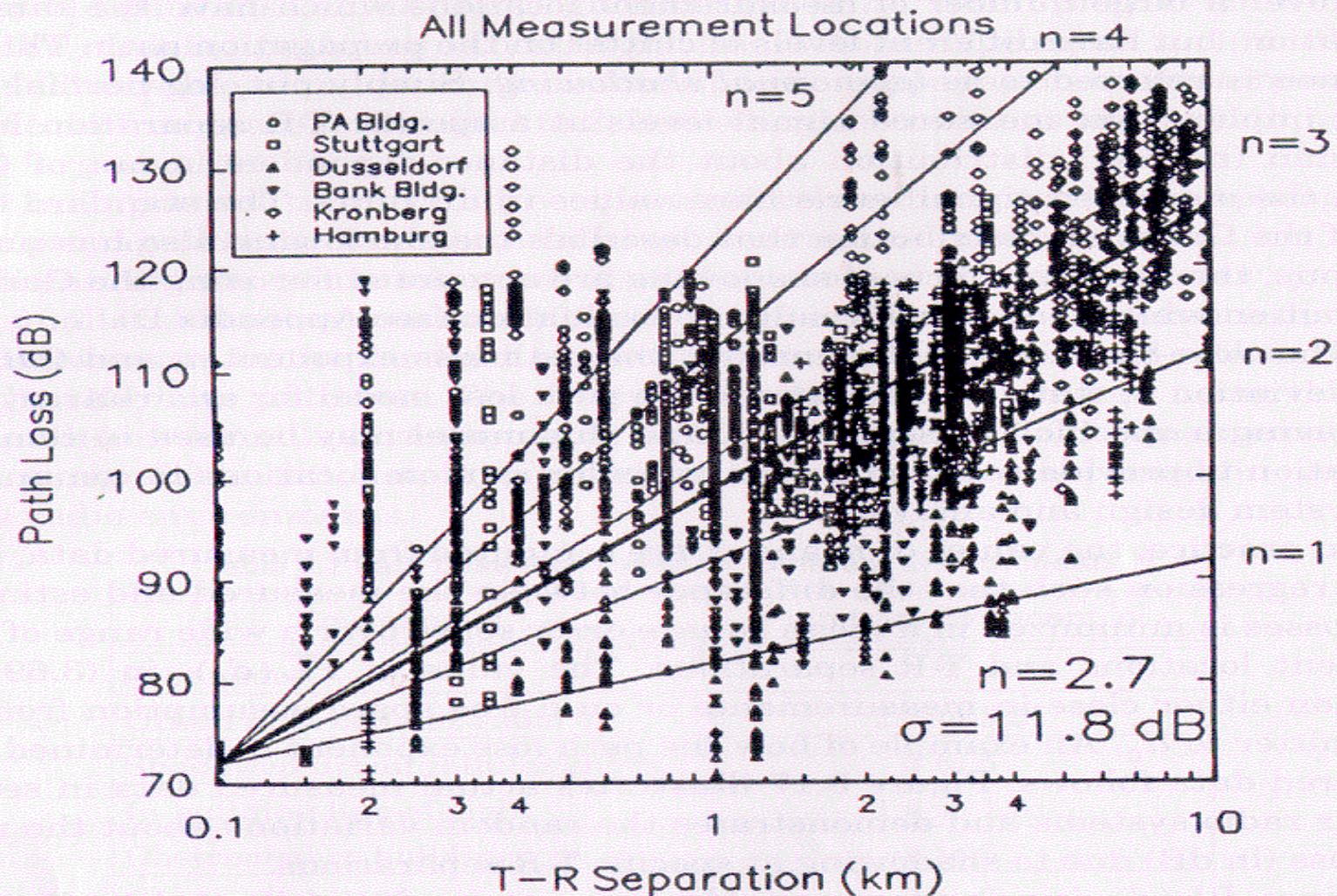


Figure 3.17

Scatter plot of measured data and corresponding MMSE path loss model for many cities in Germany. For this data, $n = 2.7$ and $\sigma = 11.8$ dB [From [Sei91] © IEEE].

Area versus Distance coverage model with shadowing model

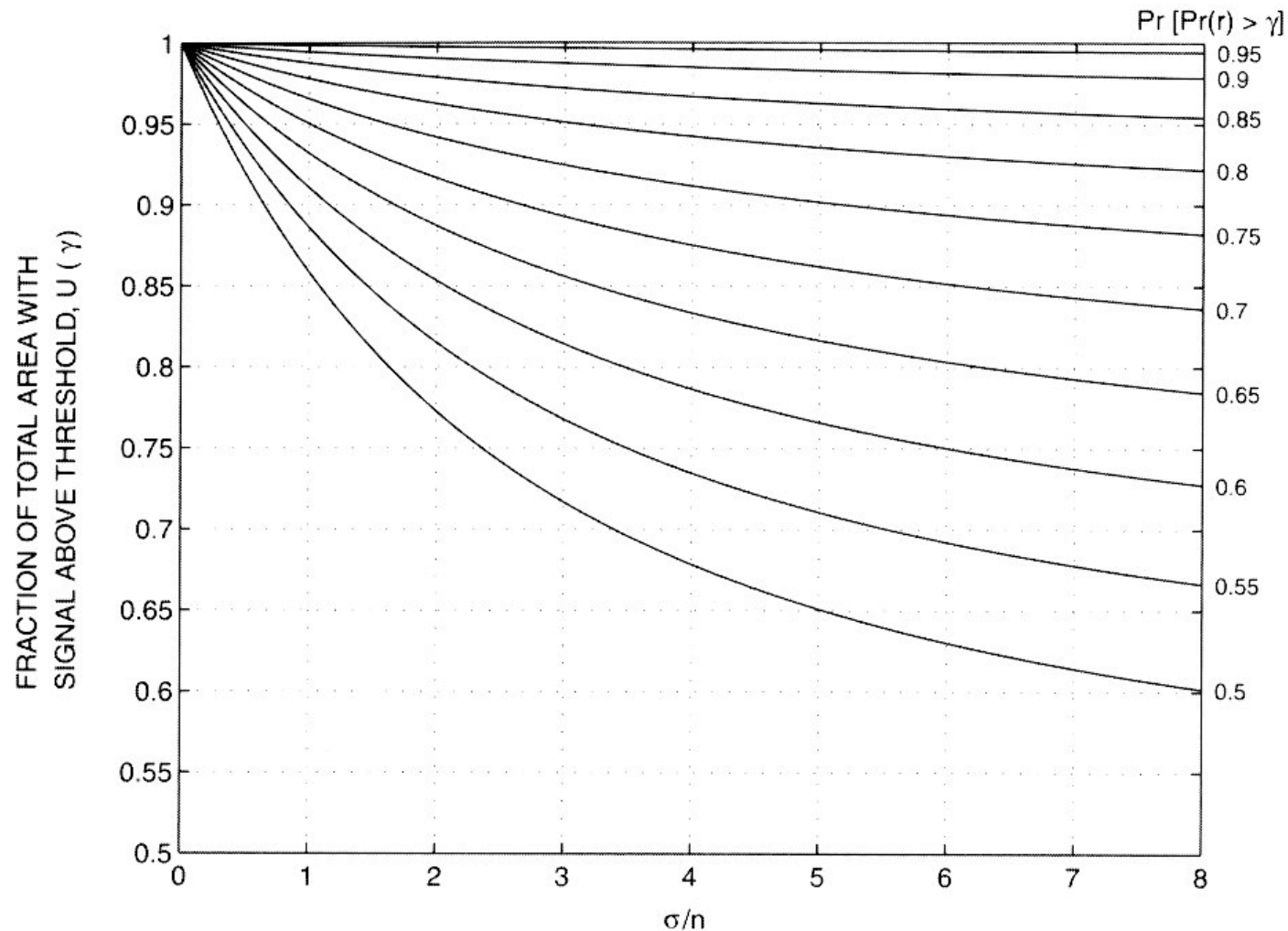
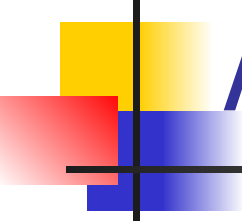


Figure 4.18 Family of curves relating fraction of total area with signal above threshold, $U(\gamma)$ as a function of probability of signal above threshold on the cell boundary.

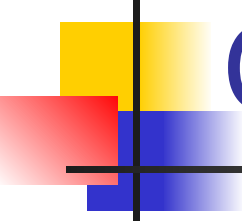


Applications

The probability that the received signal level will exceed a certain value γ can be calculated from the cumulative density function as

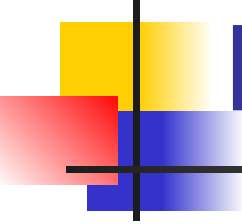
$$\Pr[P_r(d) > \gamma] = Q\left(\frac{\gamma - \overline{P_r(d)}}{\sigma}\right)$$

Can be used to determine the percentage of coverage area in cellular systems.



Outdoor Propagation Models

- There are a number of mobile radio propagation models to predict path loss over irregular terrain.
- These methods generally aim to predict the signal strength at a particular sector. But they vary widely in complexity and accuracy.
- These models are based on systematic interpretation of measurement data obtained in the service area.



Examples of Outdoor Models

- Longley-Rice Model
- **Durkin's Model**
- Okumura's Model
- Hata Model
- PCS extension to Hata Model
- Walfisch and Bertoni

2-D Propagation Raster data

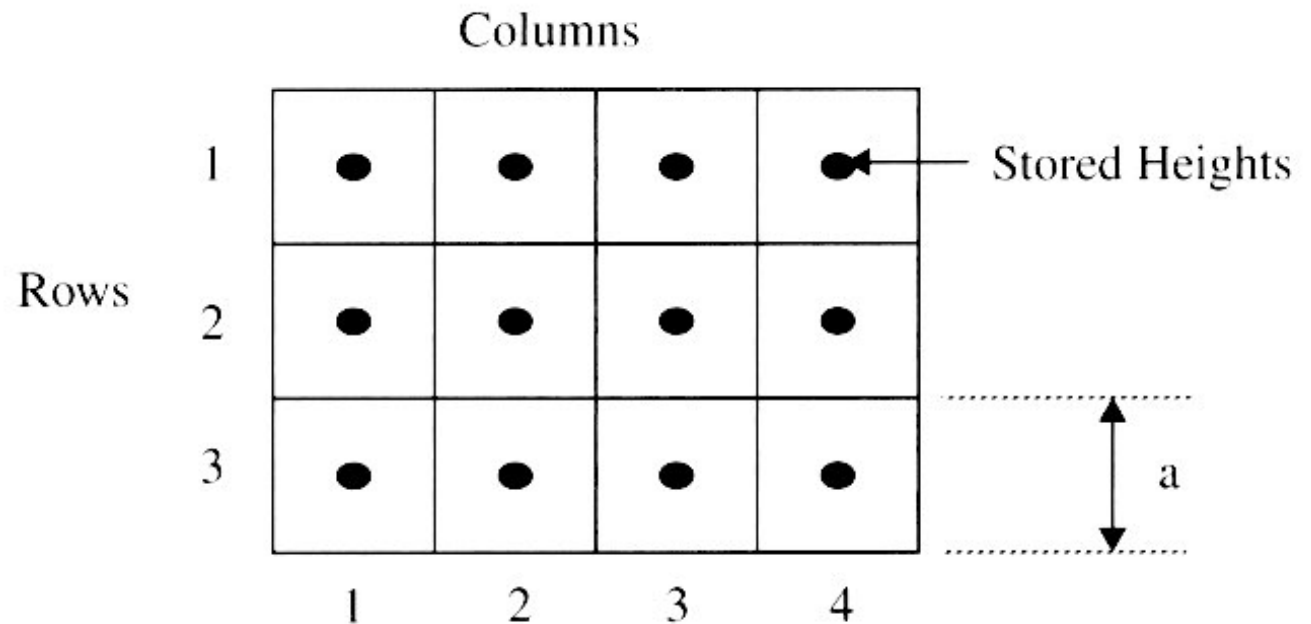
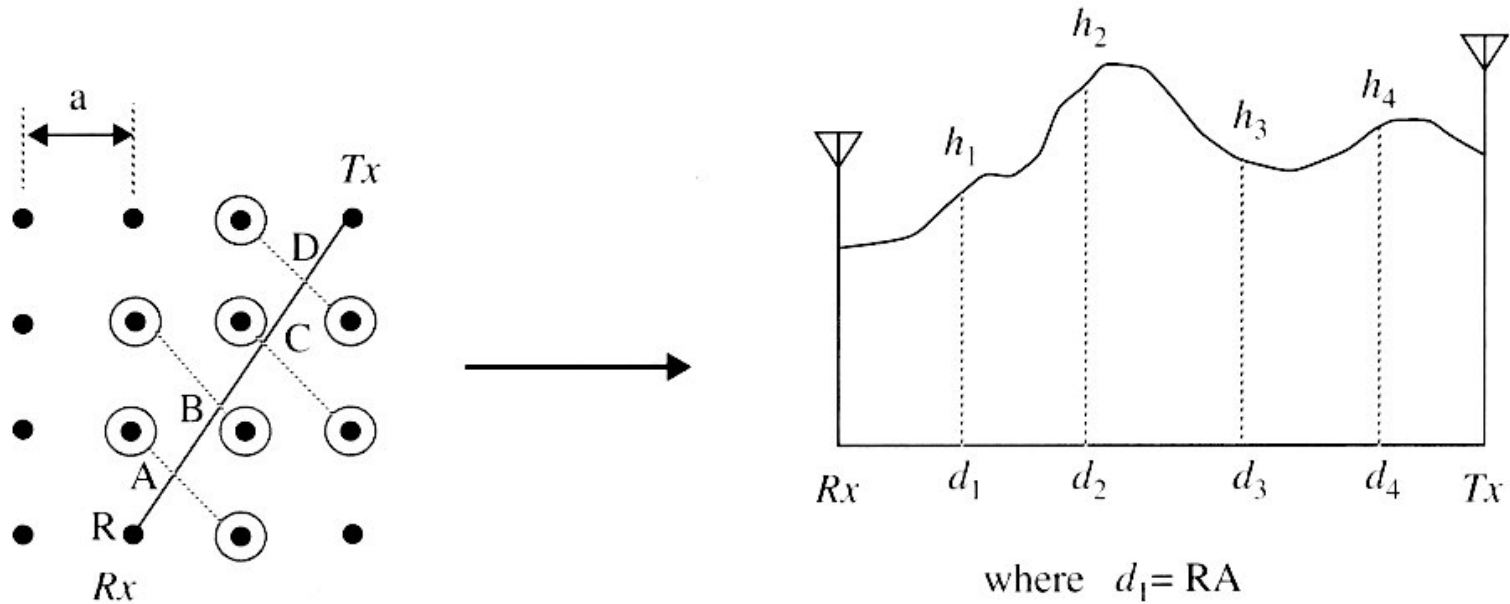


Figure 4.19 Illustration of a two-dimensional array of elevation information.

Representing propagation



(a) Top view of interpolated map
and line between Tx and Rx

(b) Side view showing reconstructed
terrain profile between Tx and Rx

Figure 4.20 Illustration of terrain profile reconstruction using diagonal interpolation.

Algorithm for line of sight (LOS)

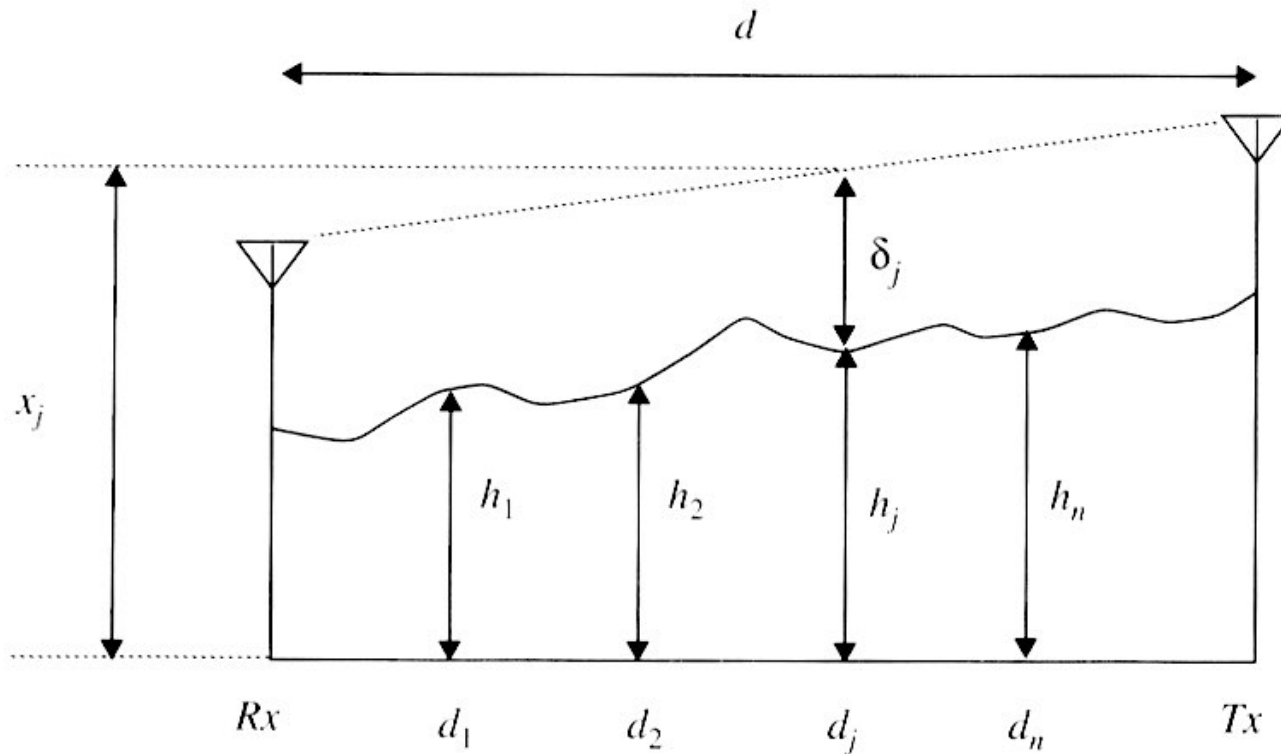
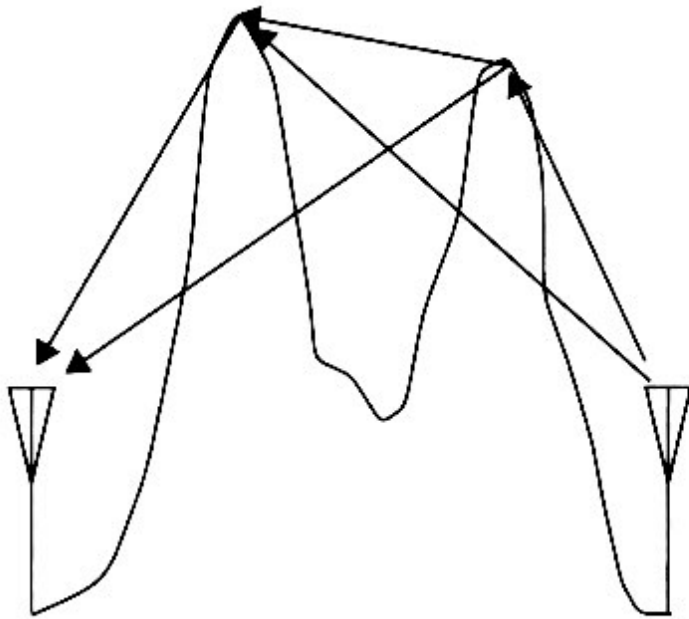
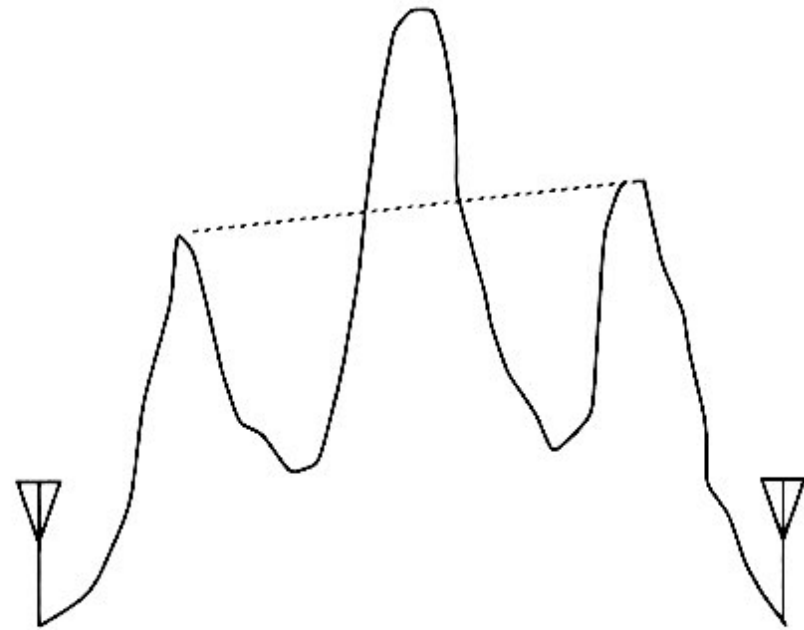


Figure 4.21 Illustration of line-of-sight (LOS) decision making process.

Multiple diffraction computation



Two Diffraction Edges



Three Diffraction Edges

Figure 4.22 Illustration of multiple diffraction edges.

Okumura and Hata's model

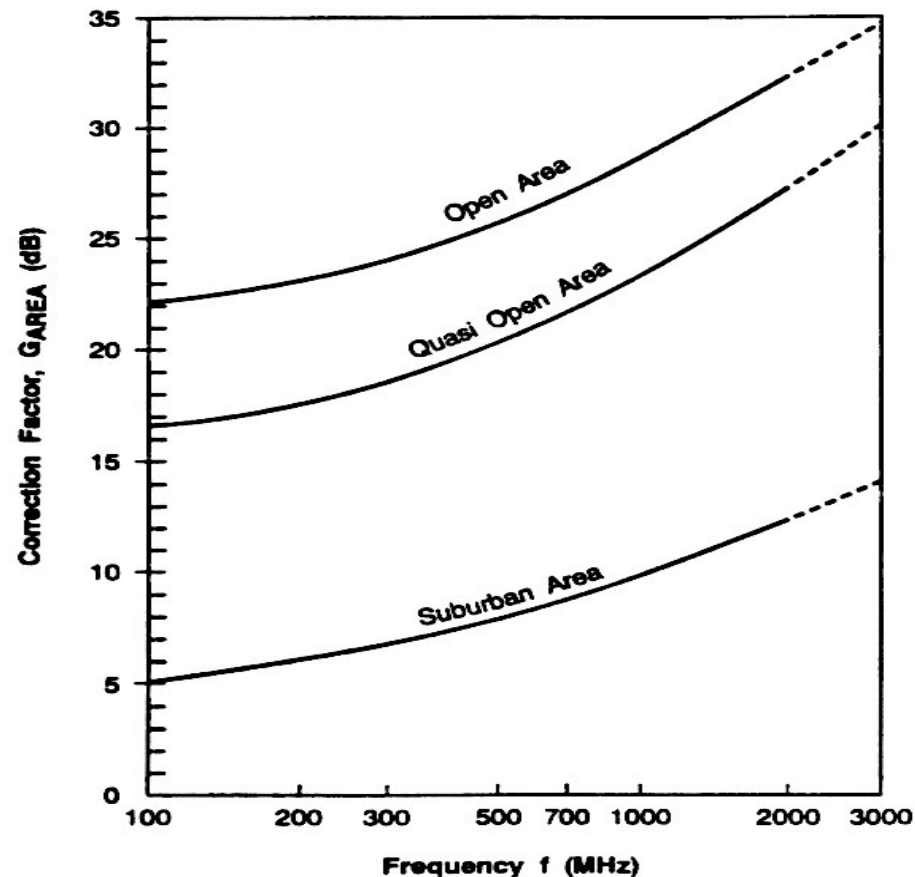


Figure 4.24 Correction factor, G_{AREA} , for different types of terrain [from [Oku68] © IEEE].

Walfisch and Bertoni's model

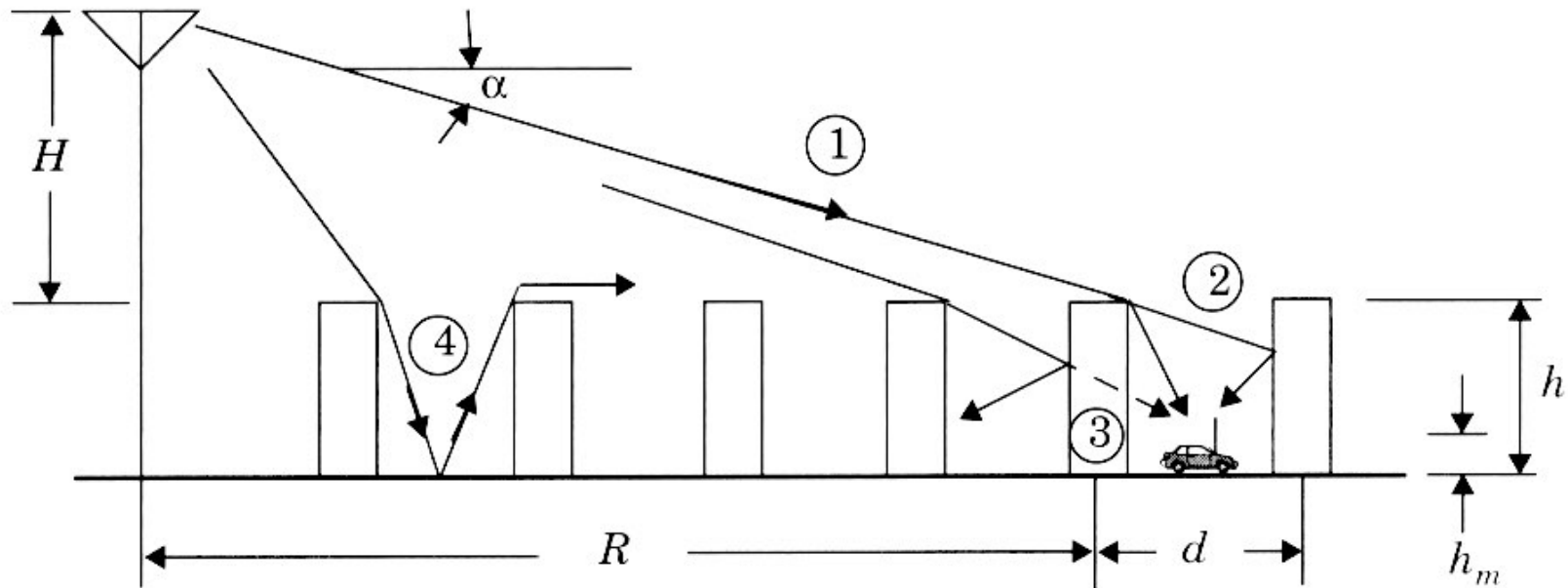
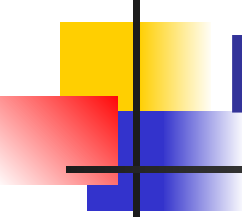


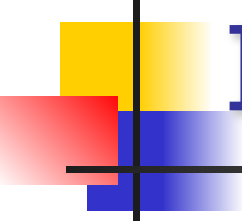
Figure 4.25 Propagation geometry for model proposed by Walfisch and Bertoni [from [Wal88] © IEEE].



Measured data from San Francisco

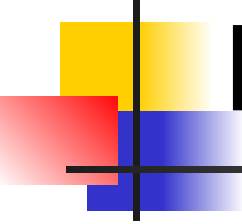
Transmitter Antenna Height	1900 MHz LOS			1900 MHz OBS	
	n_1	n_2	$\sigma(\text{dB})$	n	$\sigma(\text{dB})$
Low (3.7 m)	2.18	3.29	8.76	2.58	9.31
Medium (8.5 m)	2.17	3.36	7.88	2.56	7.67
High (13.3 m)	2.07	4.16	8.77	2.69	7.94

Figure 4.26 Parameters for the wideband microcell model at 1900 MHz [from [Feu94] © IEEE].



Indoor Propagation Models

- Indoor radio channel differs from traditional mobile radio channel in:
 - distances covered are much smaller
 - variability of the environment is greater for a much smaller range of T-R separation distances
- It is strongly influenced by specific features, such as
 - layout of the building
 - construction materials
 - building type



Log-distance Path Loss Model:

- Both theoretical and measurement-based propagation models indicate that average received signal power decreases logarithmically with distance, whether in outdoor or indoor radio channels.
- The average large-scale path loss for an arbitrary T-R separation is expressed as a function of distance by using a path loss exponent, n .

Table 3.6 Path loss exponent and standard deviation measured in different buildings [And94]

Building	Frequency (MHz)	n	σ (dB)
Retail Stores	914	2.2	8.7
Grocery Store	914	1.8	5.2
Office, hard partition	1500	3.0	7.0
Office, soft partition	900	2.4	9.6
Office, soft partition	1900	2.6	14.1
Factory LOS			
Textile/Chemical	1300	2.0	3.0
Textile/Chemical	4000	2.1	7.0
Paper/Cereals	1300	1.8	6.0
Metalworking	1300	1.6	5.8
Suburban Home			
Indoor Street	900	3.0	7.0
Factory OBS			
Textile/Chemical	4000	2.1	9.7
Metalworking	1300	3.3	6.8

Table 3.7 Path Loss Exponent and Standard Deviation for Various Types of Buildings [Sei92b]

	n	σ (dB)	Number of locations
All Buildings:			
All locations	3.14	16.3	634
Same Floor	2.76	12.9	501
Through One Floor	4.19	5.1	73
Through Two Floors	5.04	6.5	30
Through Three Floors	5.22	6.7	30
Grocery Store	1.81	5.2	89
Retail Store	2.18	8.7	137
Office Building 1:			
Entire Building	3.54	12.8	320
Same Floor	3.27	11.2	238
West Wing 5th Floor	2.68	8.1	104
Central Wing 5th Floor	4.01	4.3	118
West Wing 4th Floor	3.18	4.4	120
Office Building 2:			
Entire Building	4.33	13.3	100
Same Floor	3.25	5.2	37

Modeling Mobile Radio Small-scale fading

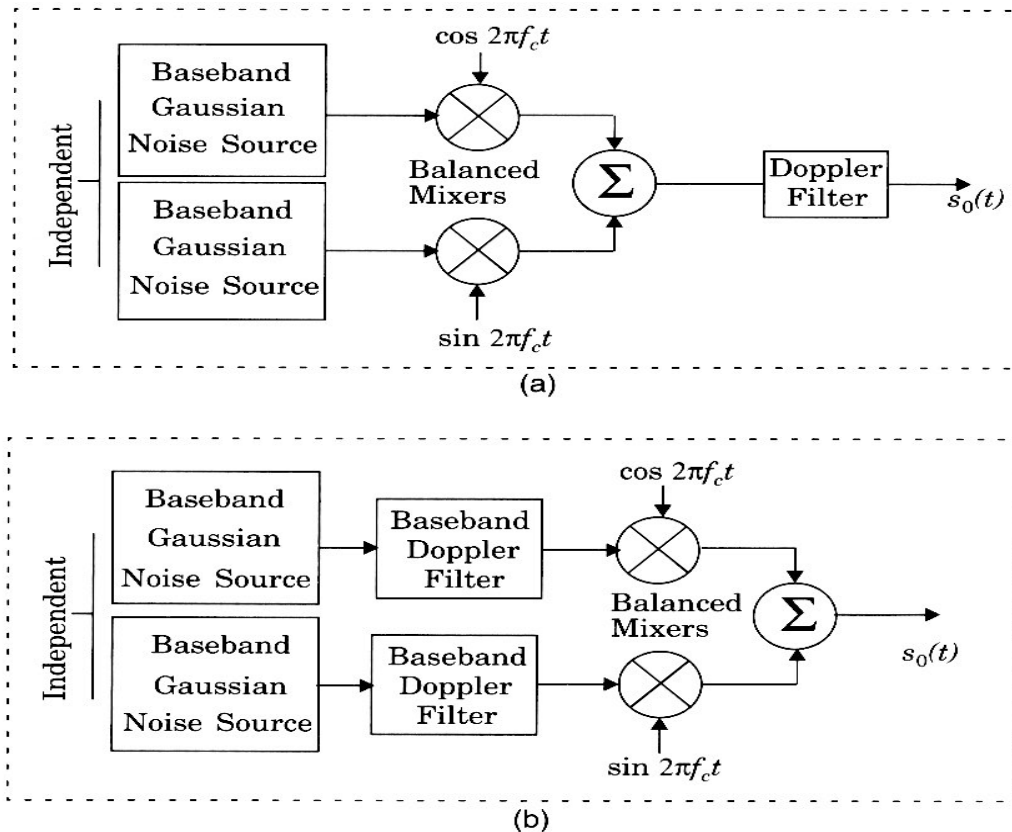


Figure 5.23 Simulator using quadrature amplitude modulation with (a) RF Doppler filter and (b) baseband Doppler filter.

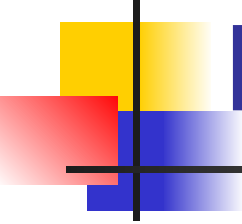
Partition losses

Table 4.3 Average Signal Loss Measurements Reported by Various Researchers for Radio Paths Obstructed by Common Building Material

Material Type	Loss (dB)	Frequency	Reference
All metal	26	815 MHz	[Cox83b]
Aluminum siding	20.4	815 MHz	[Cox83b]
Foil insulation	3.9	815 MHz	[Cox83b]
Concrete block wall	13	1300 MHz	[Rap91c]
Loss from one floor	20-30	1300 MHz	[Rap91c]
Loss from one floor and one wall	40-50	1300 MHz	[Rap91c]
Fade observed when transmitter turned a right angle corner in a corridor	10-15	1300 MHz	[Rap91c]
Light textile inventory	3-5	1300 MHz	[Rap91c]
Chain-like fenced in area 20 ft high containing tools, inventory, and people	5-12	1300 MHz	[Rap91c]
Metal blanket — 12 sq ft	4-7	1300 MHz	[Rap91c]
Metallic hoppers which hold scrap metal for recycling — 10 sq ft	3-6	1300 MHz	[Rap91c]
Small metal pole — 6" diameter	3	1300 MHz	[Rap91c]
Metal pulley system used to hoist metal inventory — 4 sq ft	6	1300 MHz	[Rap91c]
Light machinery < 10 sq ft	1-4	1300 MHz	[Rap91c]
General machinery — 10 - 20 sq ft	5-10	1300 MHz	[Rap91c]
Heavy machinery > 20 sq ft	10-12	1300 MHz	[Rap91c]
Metal catwalk/stairs	5	1300 MHz	[Rap91c]
Light textile	3-5	1300 MHz	[Rap91c]
Heavy textile inventory	8-11	1300 MHz	[Rap91c]
Area where workers inspect metal finished products for defects	3-12	1300 MHz	[Rap91c]
Metallic inventory	4-7	1300 MHz	[Rap91c]
Large I-beam — 16 - 20"	8-10	1300 MHz	[Rap91c]
Metallic inventory racks — 8 sq ft	4-9	1300 MHz	[Rap91c]
Empty cardboard inventory boxes	3-6	1300 MHz	[Rap91c]
Concrete block wall	13-20	1300 MHz	[Rap91c]
Ceiling duct	1-8	1300 MHz	[Rap91c]
2.5 m storage rack with small metal parts (loosely packed)	4-6	1300 MHz	[Rap91c]
4 m metal box storage	10-12	1300 MHz	[Rap91c]

Table 4.3 Average Signal Loss Measurements Reported by Various Researchers for Radio Paths Obstructed by Common Building Material (Continued)

Material Type	Loss (dB)	Frequency	Reference
5 m storage rack with paper products (loosely packed)	2-4	1300 MHz	[Rap91c]
5 m storage rack with large paper products (tightly packed)	6	1300 MHz	[Rap91c]
5 m storage rack with large metal parts (tightly packed)	20	1300 MHz	[Rap91c]
Typical N/C machine	8-10	1300 MHz	[Rap91c]
Semi-automated assembly line	5-7	1300 MHz	[Rap91c]
0.6 m square reinforced concrete pillar	12-14	1300 MHz	[Rap91c]
Stainless steel piping for cook-cool process	15	1300 MHz	[Rap91c]
Concrete wall	8-15	1300 MHz	[Rap91c]
Concrete floor	10	1300 MHz	[Rap91c]
Commercial absorber	38	9.6 GHz	[Vio88]
Commercial absorber	51	28.8 GHz	[Vio88]
Commercial absorber	59	57.6 GHz	[Vio88]
Sheetrock (3/8 in) — 2 sheets	2	9.6 GHz	[Vio88]
Sheetrock (3/8 in) — 2 sheets	2	28.8 GHz	[Vio88]
Sheetrock (3/8 in) — 2 sheets	5	57.6 GHz	[Vio88]
Dry plywood (3/4 in) — 1 sheet	1	9.6 GHz	[Vio88]
Dry plywood (3/4 in) — 1 sheet	4	28.8 GHz	[Vio88]
Dry plywood (3/4 in) — 1 sheet	8	57.6 GHz	[Vio88]
Dry plywood (3/4 in) — 2 sheets	4	9.6 GHz	[Vio88]
Dry plywood (3/4 in) — 2 sheets	6	28.8 GHz	[Vio88]
Dry plywood (3/4 in) — 2 sheets	14	57.6 GHz	[Vio88]
Wet plywood (3/4 in) — 1 sheet	19	9.6 GHz	[Vio88]
Wet plywood (3/4 in) — 1 sheet	32	28.8 GHz	[Vio88]
Wet plywood (3/4 in) — 1 sheet	59	57.6 GHz	[Vio88]
Wet plywood (3/4 in) — 2 sheets	39	9.6 GHz	[Vio88]
Wet plywood (3/4 in) — 2 sheets	46	28.8 GHz	[Vio88]
Wet plywood (3/4 in) — 2 sheets	57	57.6 GHz	[Vio88]
Aluminum (1/8 in) — 1 sheet	47	9.6 GHz	[Vio88]
Aluminum (1/8 in) — 1 sheet	46	28.8 GHz	[Vio88]
Aluminum (1/8 in) — 1 sheet	53	57.6 GHz	[Vio88]



Partition losses

Table 4.4 Total Floor Attenuation Factor and Standard Deviation σ (dB) for Three Buildings. Each Point Represents the Average Path Loss Over a 20λ Measurement Track [Sei92a]

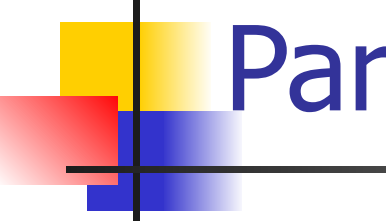
Building	915 MHz FAF (dB)	σ (dB)	Number of locations	1900 MHz FAF (dB)	σ (dB)	Number of locations
Walnut Creek						
One Floor	33.6	3.2	25	31.3	4.6	110
Two Floors	44.0	4.8	39	38.5	4.0	29
SF PacBell						
One Floor	13.2	9.2	16	26.2	10.5	21
Two Floors	18.1	8.0	10	33.4	9.9	21
Three Floors	24.0	5.6	10	35.2	5.9	20
Four Floors	27.0	6.8	10	38.4	3.4	20
Five Floors	27.1	6.3	10	46.4	3.9	17
San Ramon						
One Floor	29.1	5.8	93	35.4	6.4	74
Two Floors	36.6	6.0	81	35.6	5.9	41
Three Floors	39.6	6.0	70	35.2	3.9	27



Partition losses

Table 4.5 Average Floor Attenuation Factor in dB for One, Two, Three, and Four Floors in Two Office Buildings [Sei92b]

Building	FAF (dB)	σ (dB)	Number of locations
Office Building 1:			
Through One Floor	12.9	7.0	52
Through Two Floors	18.7	2.8	9
Through Three Floors	24.4	1.7	9
Through Four Floors	27.0	1.5	9
Office Building 2:			
Through One Floor	16.2	2.9	21
Through Two Floors	27.5	5.4	21
Through Three Floors	31.6	7.2	21



Partition losses

Table 4.6 Path Loss Exponent and Standard Deviation Measured in Different Buildings [And94]

Building	Frequency (MHz)	n	σ (dB)
Retail Stores	914	2.2	8.7
Grocery Store	914	1.8	5.2
Office, hard partition	1500	3.0	7.0
Office, soft partition	900	2.4	9.6
Office, soft partition	1900	2.6	14.1
Factory LOS			
Textile/Chemical	1300	2.0	3.0
Textile/Chemical	4000	2.1	7.0
Paper/Cereals	1300	1.8	6.0
Metalworking	1300	1.6	5.8
Suburban Home			
Indoor Street	900	3.0	7.0
Factory OBS			
Textile/Chemical	4000	2.1	9.7
Metalworking	1300	3.3	6.8

Ericsson's indoor model

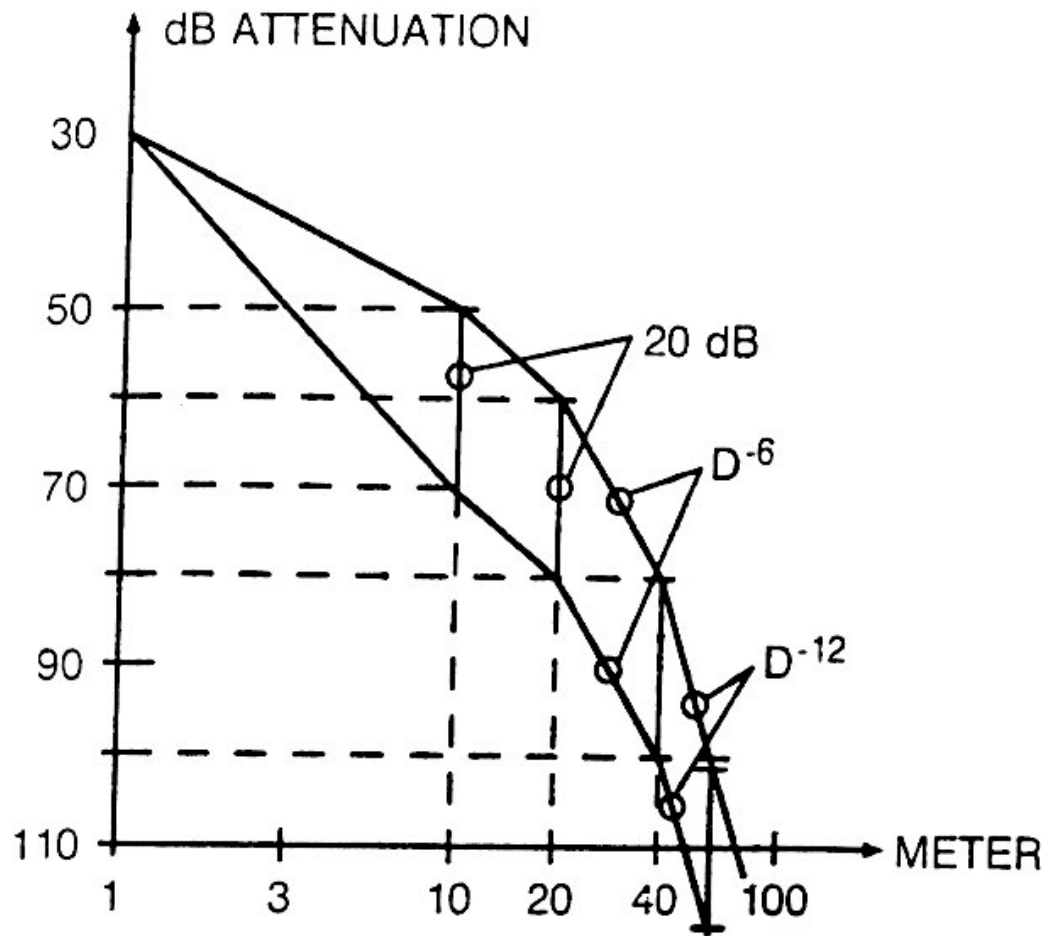


Figure 4.27 Ericsson in-building path loss model [from [Ake88] © IEEE].

Measured indoor path loss

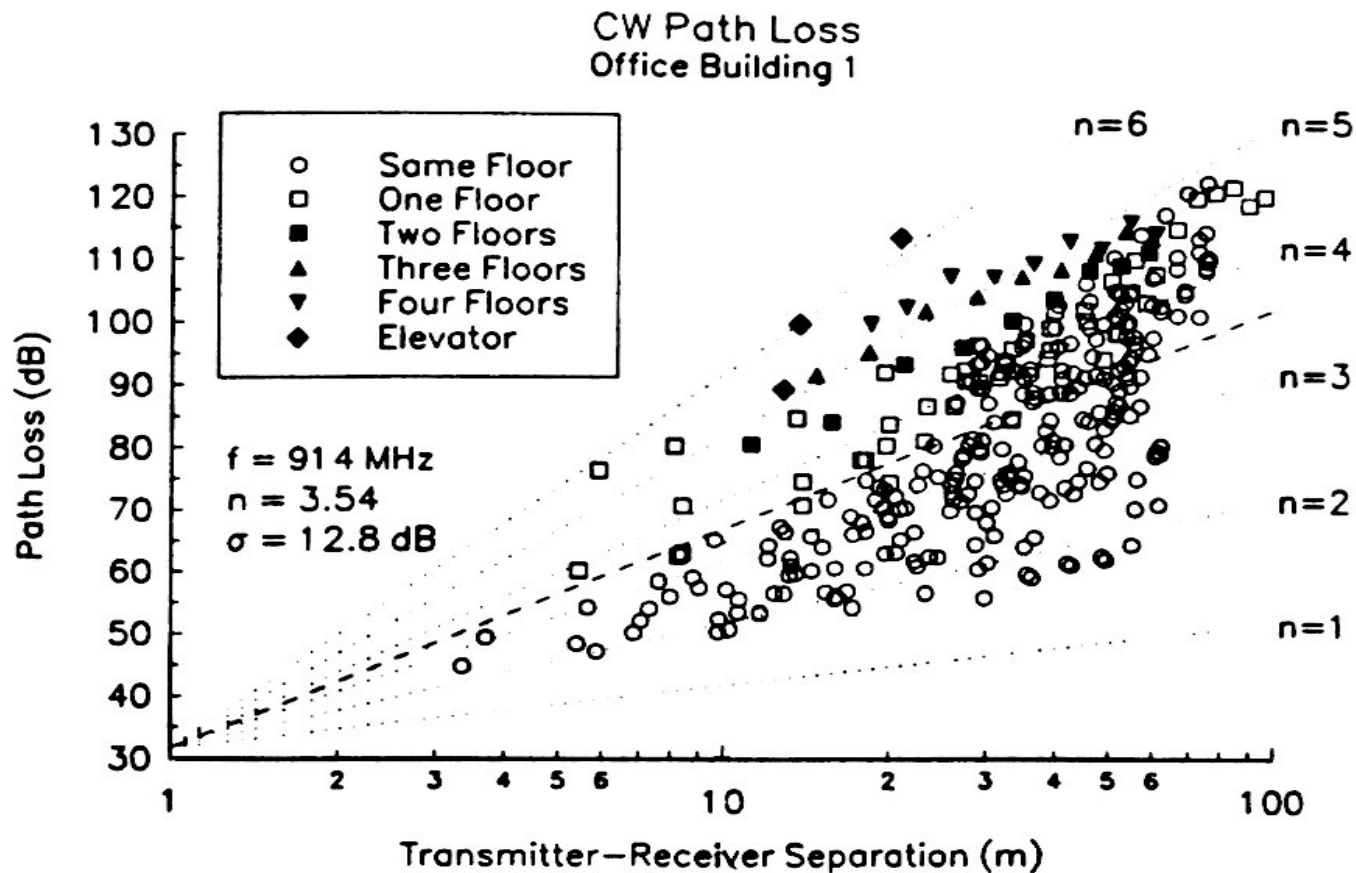


Figure 4.28 Scatter plot of path loss as a function of distance in Office Building 1 [from [Sei92b] © IEEE].

Measured indoor path loss

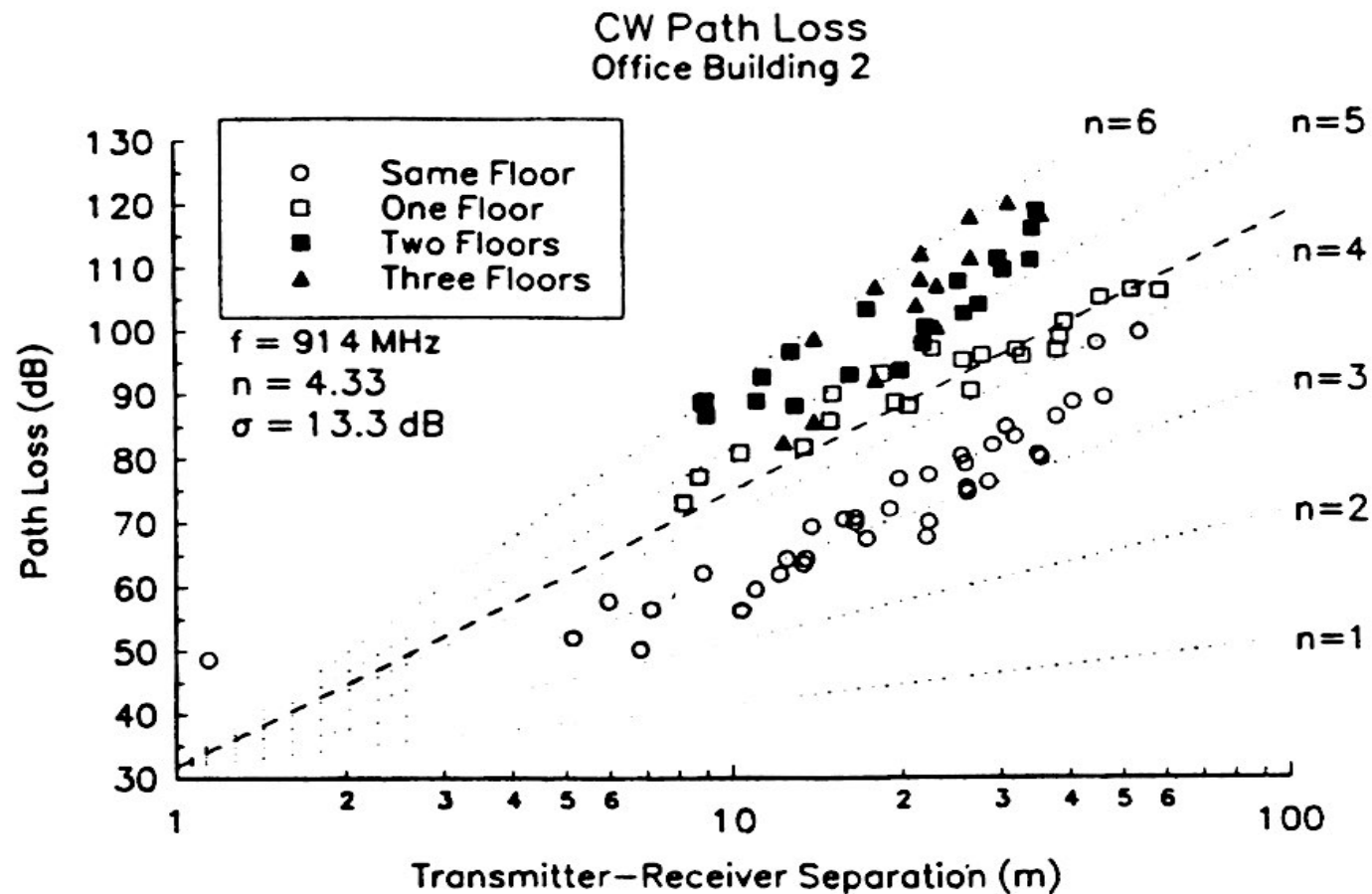
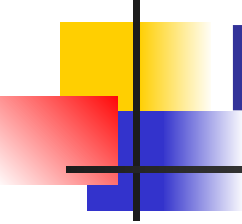


Figure 4.29 Scatter plot of path loss as a function of distance in Office Building 2 [from [Sei92b] © IEEE].



Measured indoor path loss

Table 4.7 Path Loss Exponent and Standard Deviation for Various Types of Buildings [Sei92b]

	n	σ (dB)	Number of locations
All Buildings:			
All locations	3.14	16.3	634
Same Floor	2.76	12.9	501
Through One Floor	4.19	5.1	73
Through Two Floors	5.04	6.5	30
Through Three Floors	5.22	6.7	30
Grocery Store	1.81	5.2	89
Retail Store	2.18	8.7	137
Office Building 1:			
Entire Building	3.54	12.8	320
Same Floor	3.27	11.2	238
West Wing 5th Floor	2.68	8.1	104
Central Wing 5th Floor	4.01	4.3	118
West Wing 4th Floor	3.18	4.4	120
Office Building 2:			
Entire Building	4.33	13.3	100
Same Floor	3.25	5.2	37



Devasirvatham's model

Table 4.8 Free Space Plus Linear Path Attenuation Model [Dev90b]

Location	Frequency	α -Attenuation (dB/m)
Building 1: 4 story	850 MHz	0.62
	1.7 GHz	0.57
	4.0 GHz	0.47
Building 2: 2 story	850 MHz	0.48
	1.7 GHz	0.35
	4.0 GHz	0.23

Wireless Communications

SMALL SCALE FADING





Topics:

- Parameters of Mobile Multipath Channels
- Small – Scale Fading:
 - Fading Effects due to Multipath Time Delay
 - Fading Effects due to Doppler Spread
- Rayleigh and Ricean Distributions



Major Categories of Fading :

Large Scale Fading :

This is the *loss* that propagation models try to account for mostly dependant on the distance from the transmitter to the receiver also known as *Large Scale Path Loss, Log-Normal Fading or Shadowing*

Small Scale Fading :

Could be 20-30 dB over a fraction of a wavelength. It is Caused by the superposition or cancellation of multipath propagation signals, the speed of the transmitter or receiver or the bandwidth of the transmitted signal. It is also known as *Multipath Fading or Rayleigh Fading*



Small Scale Fading:

The type of fading experienced by a signal propagating through a channel can be determined by the nature of the transmitted signal with respect to the characteristics of the channel.

Factors influencing small scale fading:

- Multipath propagation.
- Speed of the mobile.
- Speed of the surrounding objects.
- Transmission bandwidth of the signal.



Parameters of Mobile Multipath Channels:

In order to compare different multipath channels we need parameters which quantify the multipath channel, they are:

1. Delay spread
2. Coherence bandwidth
3. Doppler spread
4. Coherence time

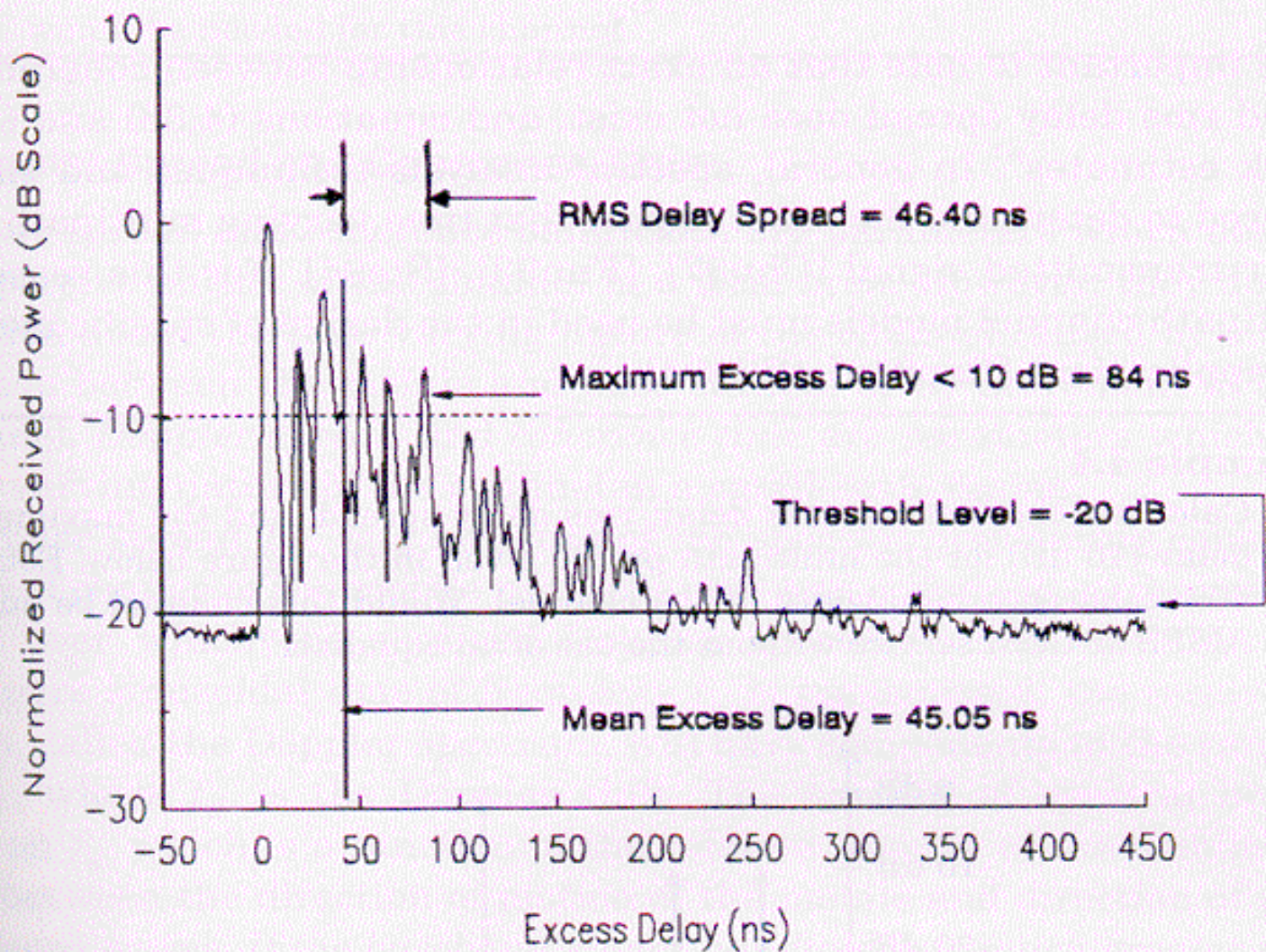


Delay Spread:

- Mean excess delay
- RMS delay spread
- Excess delay spread

Mean excess delay is the first moment of the power delay profile and is defined by the equation

$$\bar{\tau} = \frac{\sum_k a_k^2 \tau_k}{\sum_k a_k^2} = \frac{\sum_h P(\tau_k) \tau_k}{\sum_h P(\tau_k)}$$





RMS delay spread is the square root of the second central moment of the power delay profile and is defined by the equation:

$$\sigma_{\tau} = \sqrt{\overline{\tau^2} - (\bar{\tau})^2}$$

where

$$\overline{\tau^2} = \frac{\sum_k a_k^2 \tau_k^2}{\sum_k a_k^2} = \frac{\sum_h P(\tau_k) \tau_k^2}{\sum_h P(\tau_k)}$$

Maximum excess delay is defined as the $\tau_x - \tau_0$, where τ_0 is the first arriving signal and τ_x is the maximum delay at which a multipath component is within X dB of the strongest arriving multipath signal.



Coherence bandwidth:

It is the range of frequencies over which two frequency components have a potential for amplitude correlation.

If two sinusoids with a frequency separation of greater than B_c are propagating in the same channel, they are affected quite differently by the channel.



Doppler Spread and Coherence Time:

Doppler spread and Coherence Time take into account the relative motion between mobile and base station, or by movements of objects in the channel.

They describe the time varying nature of the channel in a small scale region.



Doppler Spread B_d :

When a signal of frequency f_c is transmitted, the received signal spectrum, called the Doppler spectrum, will have components $f_c - f_d$ to $f_c + f_d$, where f_d is the Doppler shift.

Coherence time T_c :

It is used to characterize the time varying nature of the frequency dispersiveness of the channel in the time domain



Small Scale Fading:

Different types of transmitted signals undergo different types of fading depending upon the relation between the

Signal Parameters: *Bandwidth, Symbol Period*
and

Channel Parameters: *RMS Delay Spread,*
Doppler Spread

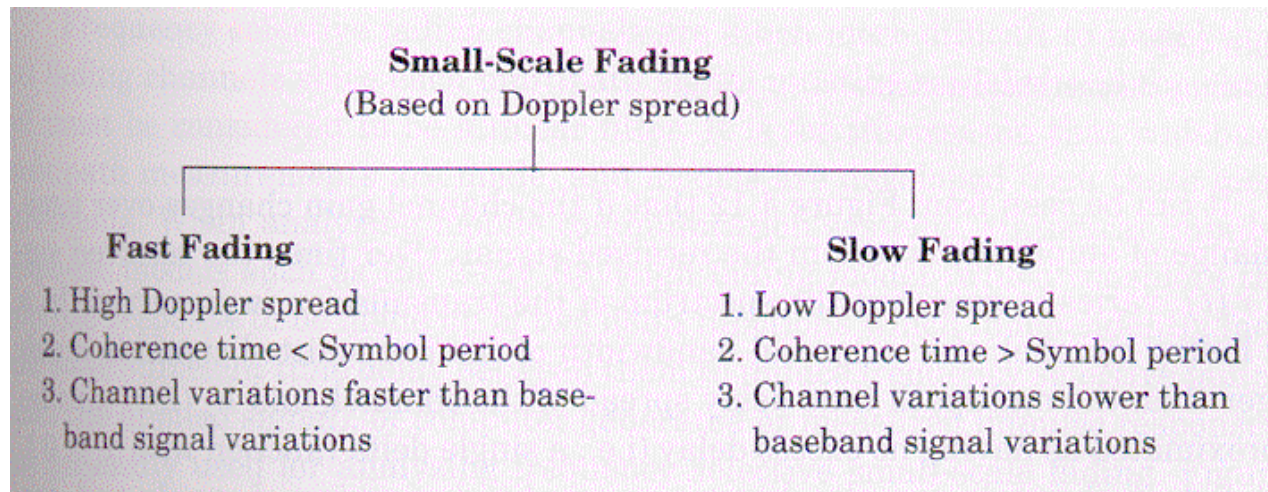
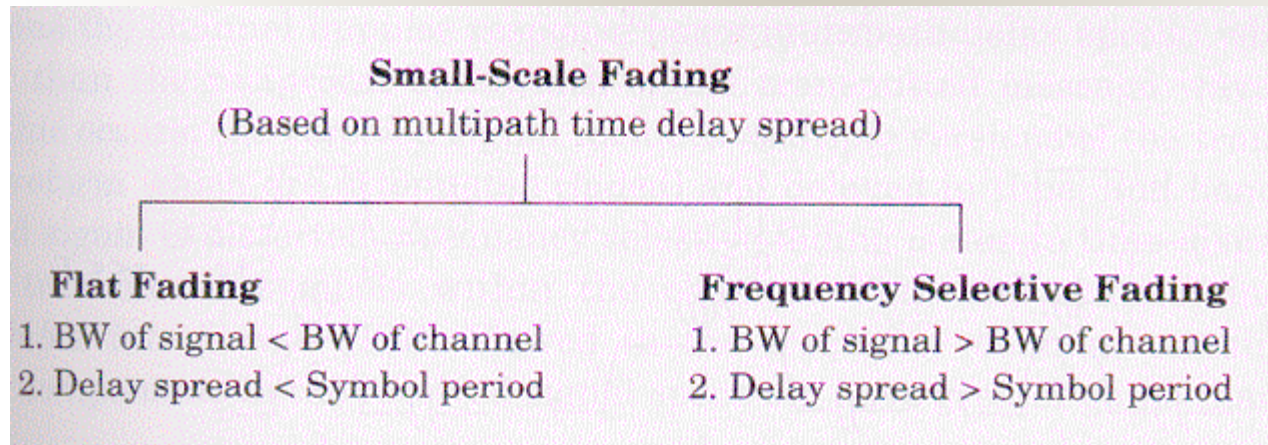
- In any mobile radio channel a wave can be dispersed either in Time or in Frequency.
- These time and frequency dispersion mechanisms lead to four possible distinct effects which depend on the nature of transmitted signal, the channel and the velocity.

Fading effects due to Multipath Time Delay Spread

- ✦ Flat Fading
- ✦ Frequency Selective Fading

Fading effects due to Doppler Spread

- ✦ Fast Fading
- ✦ Slow Fading





Flat Fading:

- A received signal is said to have underwent Flat Fading if “*The Mobile Radio Channel has a constant gain and linear phase response over a Bandwidth which is greater than the Bandwidth of the transmitted Signal*”
- Fading in which all frequency components of a received radio signal vary in the same proportion simultaneously

- Here the multipath structure of the channel is such that spectral characteristics of the transmitted signal are preserved at the receiver
- But due to the fluctuations in the gain of the channel caused by multipath, the signal strength varies with time

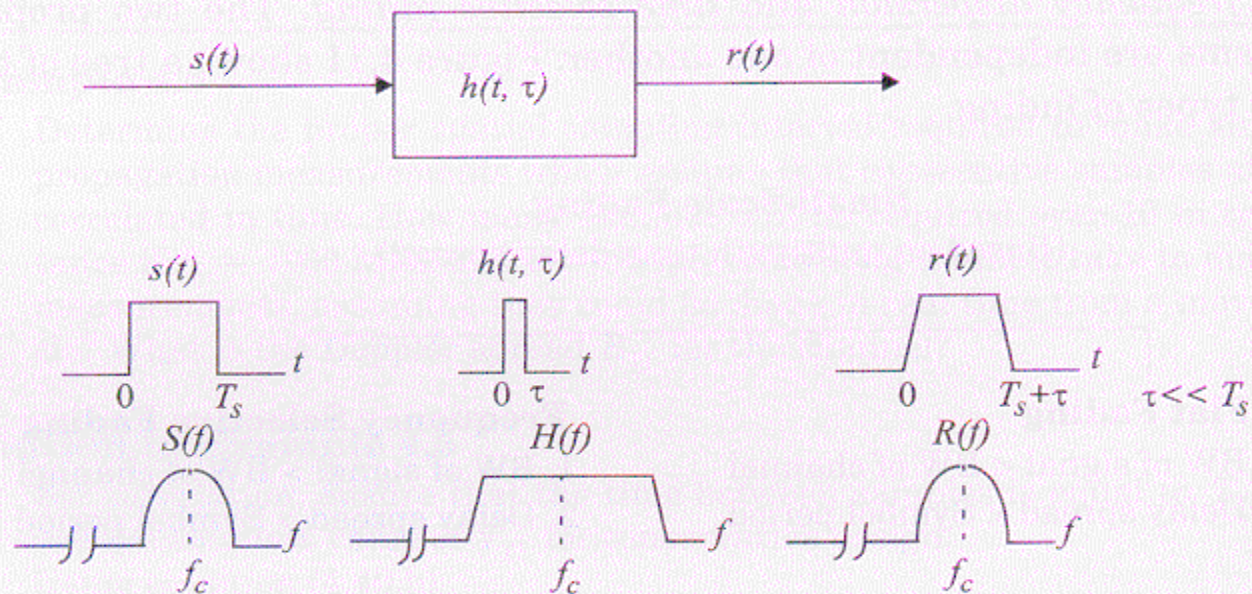


Figure 4.12
Flat fading channel characteristics.

- 
- From the figure we can note that if the channel gain varies with time, a change of amplitude of the received signal occurs.
 - From the figure we can note that the spectrum of the received signal $r(t)$ is preserved even though there is a change in gain.

Flat fading channels are also referred as amplitude varying channels or narrow band channels, since the bandwidth of the applied signal is narrow as compared to the channel flat fading bandwidth.

- 
- Typical Flat fading channels cause deep fades
 - To achieve low bit error rates during times of deep fades, Flat fading channels operate at 20 to 30dB more transmitter power compared to the systems operating over non-fading channels.
 - For designing Radio links, the distribution of the instantaneous gain of flat fading channels is important.
 - Rayleigh distribution is the most common amplitude distribution.
 - According to this distribution, Rayleigh Flat fading channel model assumes that the channel induces an amplitude which varies in time.



Summary

Signal undergoes Flat Fading if:

- $B_s \ll B_c$

where

B_s is bandwidth and

B_c is the coherence bandwidth of the channel

And

- $T_s \gg \sigma_\tau$

where

T_s is the reciprocal bandwidth and

σ_τ rms delay spread.



Frequency Selective Fading:

The channel creates frequency selective fading on the received signal when the channel *possesses a constant gain and linear phase response over a bandwidth, which is smaller than the bandwidth of the transmitted signal*

- Under these conditions the channel impulse response has a multipath delay spread which is greater than the reciprocal bandwidth of the transmitted message waveform
- So the received signal includes multiple versions of the transmitted waveform, which are attenuated and delayed in time, and hence the received signal is distorted.

- 
- Frequency selective fading is much difficult to model than flat fading channels because each multipath signal must be modeled and the channel must be considered to be a linear filter
 - It is for this reason that wideband multipath measurements are made and models are developed from these measurements
 - When analyzing mobile communication systems, statistical impulse response models such as the 2-ray Rayleigh model or computer generated or measured impulse responses are generally used for analyzing frequency selective small-scale fading.

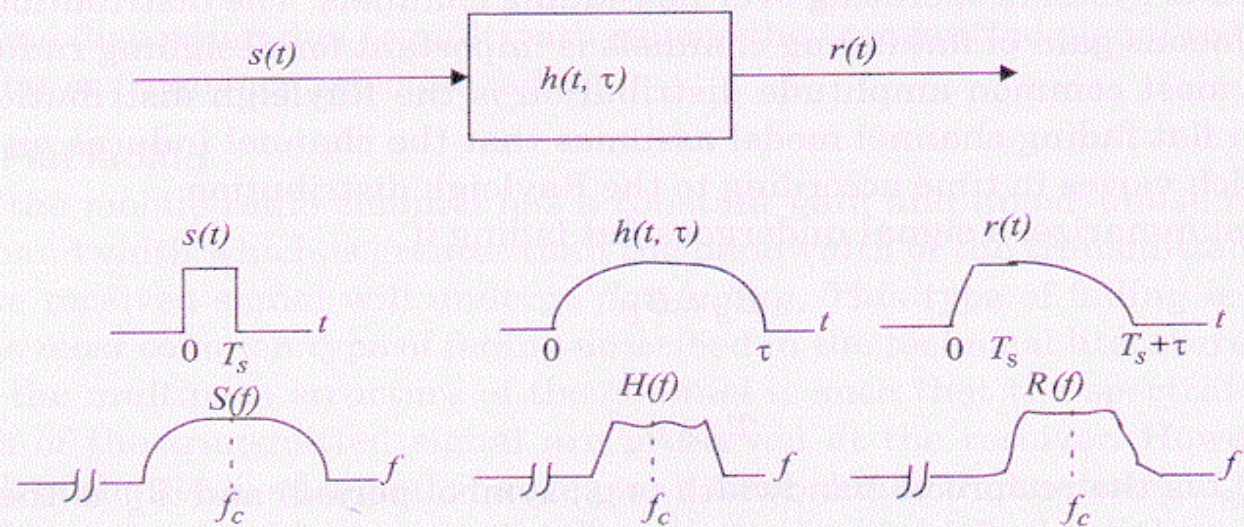


Figure 4.13
Frequency selective fading channel characteristics.

- 
- For frequency selective fading, the spectrum $S(f)$ of the transmitted signal has a bandwidth which is greater than the coherence bandwidth B_c of the channel
 - Frequency selective fading is caused by multipath delays which approach or exceed the symbol period of the transmitted symbol
 - These channels are also known as wideband channels since the bandwidth of the signal $s(t)$ is wider than the bandwidth of the channel impulse response
 - As time varies, the channel varies in gain and phase across the spectrum of $s(t)$, resulting in time varying distortion in the received signal $r(t)$



Summary

Signal undergoes Frequency Selective Fading if:

- $B_s > B_c$

where

B_s is bandwidth and

B_c is the coherence bandwidth of the channel

And

- $T_s < \sigma_\tau$

where

T_s is the reciprocal bandwidth and

σ_τ rms delay spread.



Fading Effects Due to Doppler Spread:

- Channels are also classified depending upon how rapidly the transmitted baseband signal changes compared to the rate of change of channel.
- It is either Fast Fading or Slow Fading channel.
- The velocity of the mobile (or the velocity of the objects in the channels) and the baseband signaling determines whether a signal undergoes fast or slow fading.



Fast Fading:

- In Fast Fading channel the channel impulse response changes at a rate much faster than the transmitted baseband signal.
- In other words the coherence time of the channel is smaller than the symbol period of the transmitted signal.
- This causes frequency dispersion due to Doppler spreading, which leads to signal distortion.
- When viewed in frequency domain, signal distortion due to fast fading increases with increasing Doppler spread relative to the bandwidth of the transmitted signal.



Hence a signal will undergo fast fading if

$$T_s > T_c$$

and

$$B_s < B_D$$

Note: *When a channel is specified as Fast or Slow fading channel, it does not specify whether the channel is flat fading or frequency selective in nature*



- In the case of flat fading channel, we can approximate the impulse response to be a delta function without the time delay.

- Hence, a flat fading, fast fading channel is a channel in which the amplitude of the delta function varies faster than the rate of change of the baseband transmitted signal
- A frequency selective, fast fading channel, the amplitudes, phases, and the time delays of any one of the multipath components vary faster than the rate of change of transmitted signal
- In practice, fast fading occurs only for very low data rates



Slow Fading:

- In Slow Fading channel the channel impulse response changes at a rate much slower than the transmitted baseband signal
- Here the channel may be assumed static over one or several bandwidth intervals.
- In the frequency domain, this implies that the Doppler spread of the channel is much less than the bandwidth of the baseband channel.



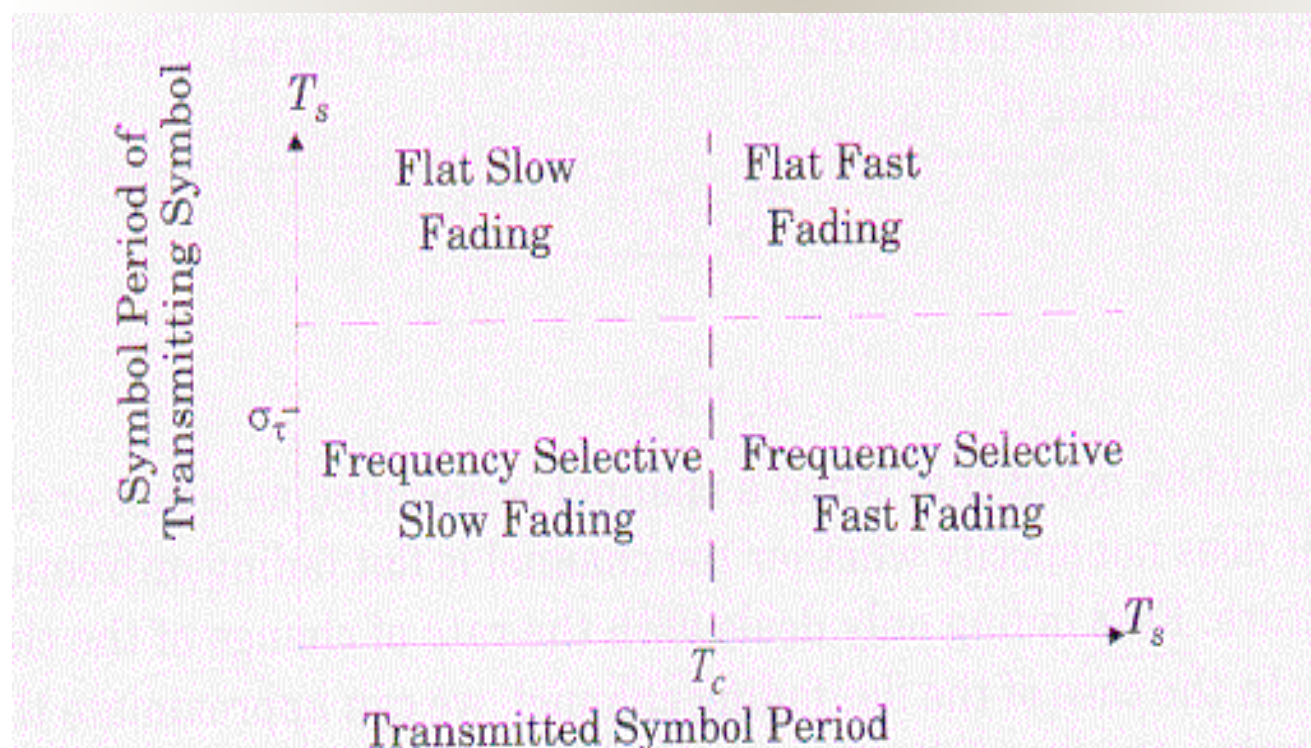
Hence a signal will undergo slow fading if

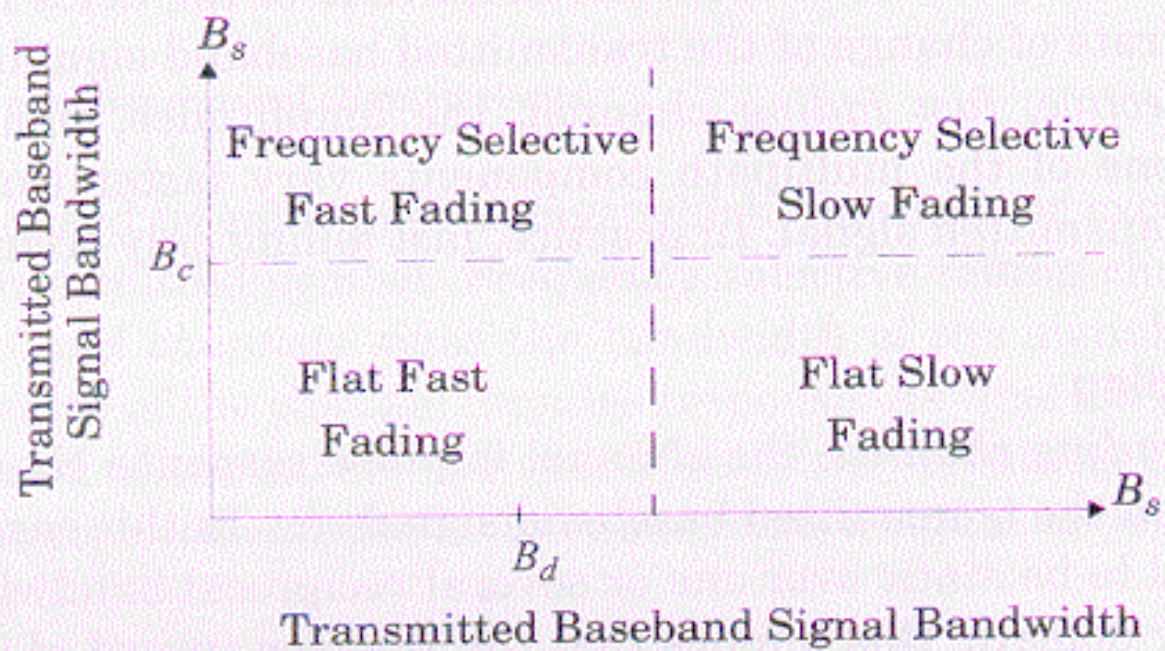
$$T_s \ll T_c$$

and

$$B_s \gg B_D$$

Note: *Fast and Slow Fading deal with the relationship between the time rate of change in the channel and the transmitted signal, and not with the propagation path loss models.*







Rayleigh Fading Distribution:

Rayleigh Fading Distribution in mobile radio channels is commonly used to describe the *statistical time varying nature* of the received envelope of a flat fading signal or the envelope of an individual multipath component.

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & (0 \leq r \leq \infty) \\ 0 & (r < 0) \end{cases}$$

where

σ is the rms value of the received voltage signal before envelope detection,

σ^2 is the time-average power of the received signal before envelope detection.

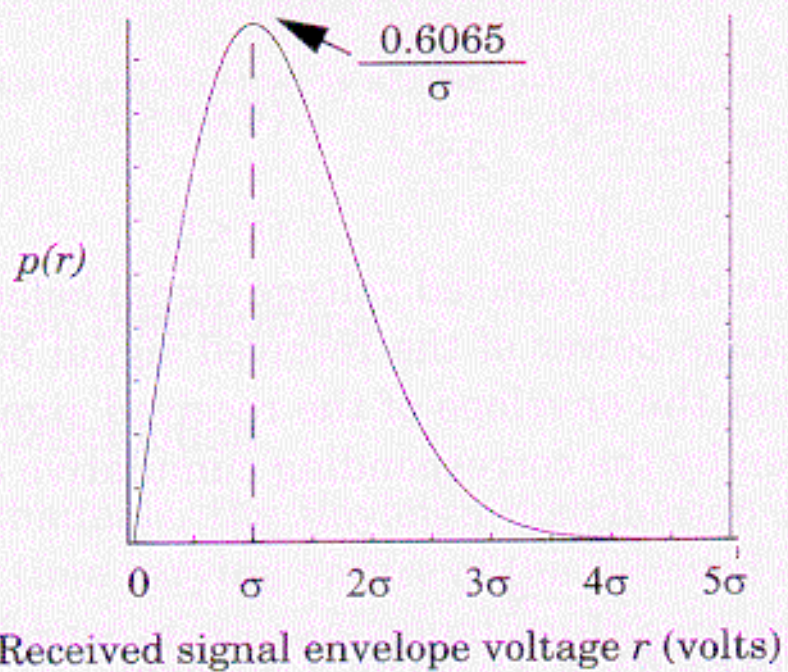


Figure 4.16
Rayleigh probability density function (pdf).



The probability that the envelope of the received signal does not exceed a specified value R is given by the corresponding cumulative distribution function(CDF)

$$P(R) = P_r(r \leq R) = \int_0^R p(r)dr = 1 - \exp\left(-\frac{r^2}{2\sigma^2}\right)$$

The value r_{mean} of the Rayleigh distribution is given by

$$r_{mean} = E[r] = \int_0^{\infty} rp(r)dr = \sigma \sqrt{\frac{\pi}{2}} = 1.2533\sigma$$



The variance σ_r^2 of the Rayleigh distribution is given by

$$\begin{aligned}\sigma_r^2 &= E[r^2] - E^2[r] = \int_0^{\infty} r^2 p(r) dr - \frac{\sigma^2 \pi}{2} \\ &= \sigma^2 \left(2 - \frac{\pi}{2} \right) = 0.4292 \sigma^2\end{aligned}$$

which represents the ac power in the signal envelope.

The rms value of the envelope is $\sqrt{2}\sigma$



The median value of r is found by solving

$$\frac{1}{2} = \int_0^{r_{median}} p(r) dr$$

$$r_{median} = 1.177\sigma$$

Note:

- *It is customary to use median values instead of the mean values, since fading data are usually measured in the field and a particular distribution cannot be assumed.*
- *By using median values instead of mean values it is easier to compare different fading distributions which have widely varying means*



Ricean Fading Distribution:

- When there is a *dominant stationary* (nonfading) signal component present, such as a line-of-sight propagation path, the small scale-scale fading envelope distribution is Ricean.
- Random multipath components arriving at different angles are superimposed on a stationary dominant signal
- At the output of an envelope detector this has the effect of adding a dc component to the random multipath

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- The effect of a dominant signal arriving with many weaker multipath signals gives rise to Ricean distribution
 - As the dominant signal becomes weaker, the composite signal gives resembles a noise signal which has an envelope that is Rayleigh
 - Thus, the Ricean distribution degenerates to a Rayleigh distribution when the dominant component fades away

- The Ricean distribution is given by

$$p(r) = \begin{cases} \frac{r}{\sigma^2} e^{-\frac{(r^2 + A^2)}{2\sigma^2}} I_0\left(\frac{Ar}{\sigma^2}\right) & \text{For } A \geq 0, r \geq 0 \\ 0 & \text{For } r < 0 \end{cases}$$

Where

A denotes peak amplitude of the dominant signal
 $I_0(\cdot)$ is the modified Bessel function of the first
kind and zero order

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- The Ricean distribution is often defined in terms of a parameter K called the Ricean Factor

$$K = \frac{A^2}{2\sigma^2}$$

Where K is defined as the ratio between the deterministic signal power and the variance of the multipath.

$$K(dB) = 10 \log \frac{A^2}{2\sigma^2} dB$$

- Ricean Factor K completely determines the Ricean distribution.
- As $A \rightarrow 0$, $K \rightarrow -\infty$ dB, and as the dominant path decreases in amplitude, the Ricean distribution degenerates to Rayleigh distribution

